



고려대학교 약학대학
KOREA UNIVERSITY COLLEGE OF PHARMACY

KU-The Future
110 years and beyond

Microbiology



Fall semester, 2026

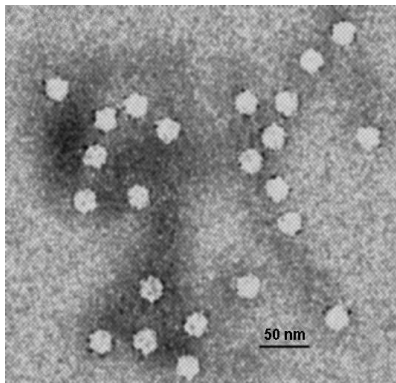
► Basic information of virus

- What is...?

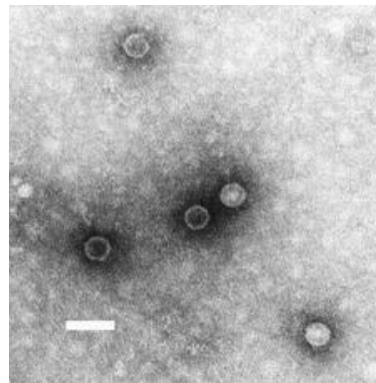
- : Small particle that can infect other organisms
 - Small infectious agent (filterable)

- How small is...?

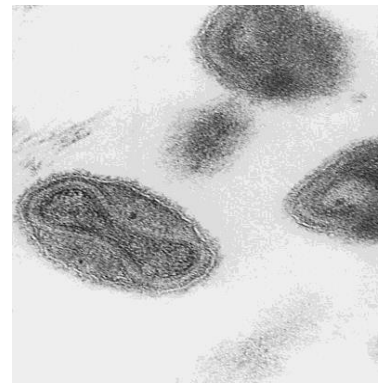
- : 20 ~ 400 nm (900 nm)
 - too small to be seen directly with a light microscope
 - can be seen by an electron microscope (ultramicroscopic)



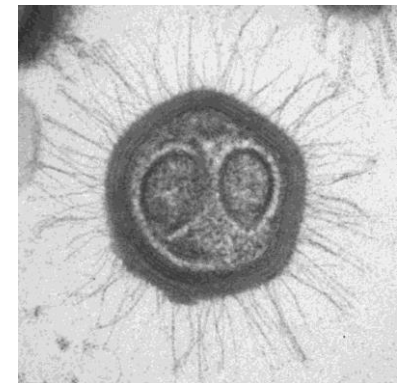
Parvovirus
(20 ~ 30 nm)



Poliovirus
(20 ~ 30 nm)



Smallpox
(350 ~ 400 nm)



Mimivirus
(400 nm; 900 nm)

► Basic information of virus

- Whom virus infect ?
 - all types of organisms: from eukaryotes to bacteria (including Archaea)



Smallpox



FMD



TMV



Bacteriophage



Archaeal virus

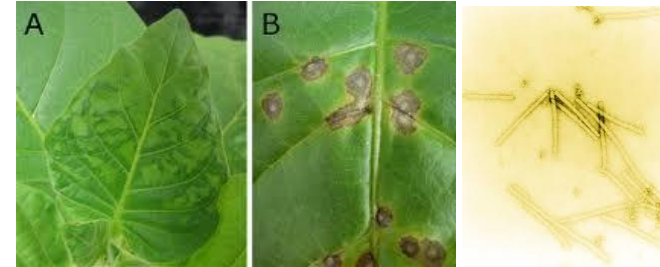
- How many type...?
 - millions of different types / 5,000 described in detail
 - the most abundant type of biological entity

► Basic information of virus

- Etymology :
 - the Latin *virus* : meaning “poison”
- History :
 - **Edward Jenner**: jennerization (종두법) – smallpox (천연두) vaccination / prevention of smallpox using vaccinia virus (우두바이러스)
 - **Louis Pasteur**: developed rabies (광견병) vaccine by using the weakened form of the disease organism
 - **Charles Chamberland**: invented a filter having a pore size smaller than bacteria (called “Chamberland filter”)
 - **Adolf Mayer**: first discovered and named “mosaic disease of tobacco (TMD)” → demonstrating the infectivity of the sap from the affected tobacco plants

► Basic information of virus

- History :
 - **Demitry Ivanovsky**: a causative agent for TMD – filterable / toxin produced by bacteria
 - **Martinus Beijerinck**: a causative agent for TMD – “soluble living germ”
→ virus being liquid (not true)
 - **Wendell Stanley**: isolation of a nucleoprotein displaying TMD activity
→ virus being particulate (true)
 - **Friedrich Loeffler**: realized the first animal virus (foot-and-mouth disease)
 - **20 the century**: bacteriophage discovery, influenza and polio virus cultures, electron microscope, reverse transcriptase, etc.



► Basic information of virus

- **Where did virus come from?**

- not clear, but together with living cell's evolution
- hypotheses to explain the origin:

1. **progressive hypothesis (escape or vagrancy hypothesis):**

- being originated via a progressive process of genetic materials capable of moving within a genome (ex: plasmids and jumping genes)
 - gaining the ability to exit one cell and enter another
- (vs.) cannot explain the existence of viral proteins

2. **regressive hypothesis (reduction or degeneracy hypothesis):**

- originated via a regressive process of obligate intracellular parasitic bacteria (e.g., *Rickettsia* and *Chlamydia* species) → loss of genes related to self-replication
- (vs.) the smallest parasites not resembling viruses

3. **virus-first hypothesis (coevolution hypothesis):**

- existence of virus before cells (in a pre-cellular world as a self-replicating unit)
 - parallel evolution of viruses and cellular organisms
- (vs.) contravening requirement of host cells for replication

► Basic information of virus

- Living vs. Non-living organisms

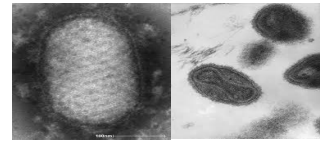
- properties as living organisms:

1. possessing own genes (DNA or RNA)
2. evolution
3. reproduction
4. self-assembly

- properties as non-living organisms:

1. need hosts for reproduction (intracellular parasites)
2. not having own metabolism
3. not having a cellular structure (e.g. no ribosome, no mitochondria, etc.)

→ basic units of life; organisms at the edge of life



Poxvirus



► Basic information...

- Terms in virology

- **capsid**: protein shell of a virus

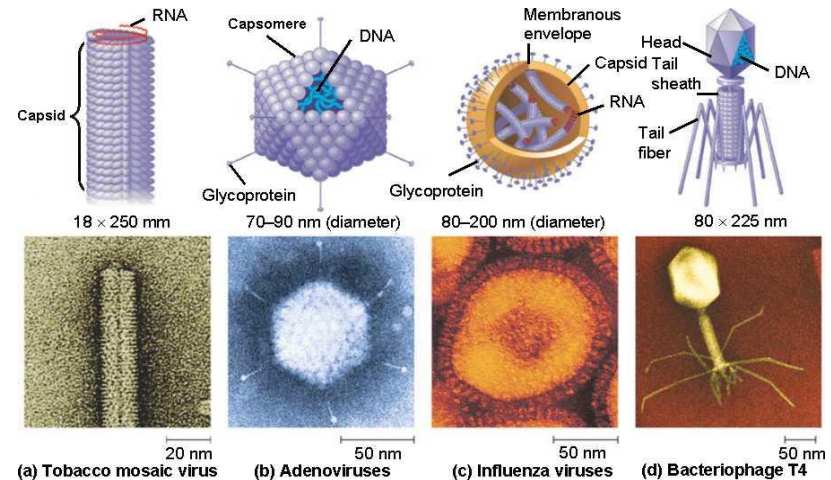
- classified according to the structure: icosahedral, helical, more complicated (e.g. bacteriophages), etc.
 - helical shape: cylindrical (e.g. tobacco mosaic virus)
 - icosahedral shape: 20 equilateral triangular faces, approximates a sphere

- **capsomer**: a basic subunit of the capsid / an outer covering of protein that protects the genetic material of a virus

→ capsomers self-assemble to form the capsid

- **protomer**: a subunit of the capsomer

→ protomers aggregate to form the capsomer (5–6 protomers)



► Basic information...

- Terms in virology

- **lipovirus**: virus wrapped in a membranous envelope

- viral envelope:

- derived from portions of the host cell membranes
 - components: phospholipids + proteins (viral glycoproteins)
 - function: helps viruses enter host cells
 - sensitive to desiccation, heat, detergents, and fat-solubles (e.g. ether, chloroform, bile salts)

- easier to sterilize than non-enveloped viruses
 - have limited survival outside host environments
 - must transfer directly from host to host
 - possess great adaptability and can change rapidly in a short time to survive
 - can cause persistent infections

► Basic information...

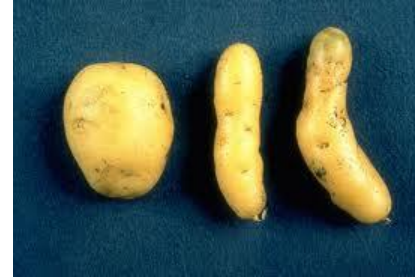
• Terms in virology

– viroid:

- In 1971, first discovered by T. O. Diener
- a small circular RNA molecule causing plant diseases
 - ex: potato spindle-tuber diseases, coconut cadang-cadang, etc.
 - a few hundred nucleobases: 240–600 nucleotides
 - lacking a protein coat
 - interfering with the cell's normal path of gene expression
 - RNA silencing

– provirus:

- virus genome that is integrated into the DNA of a host cell (ex: retrovirus)
- (+) ss RNA → (–) DNA → (±) DNA → (+) mRNA → protein



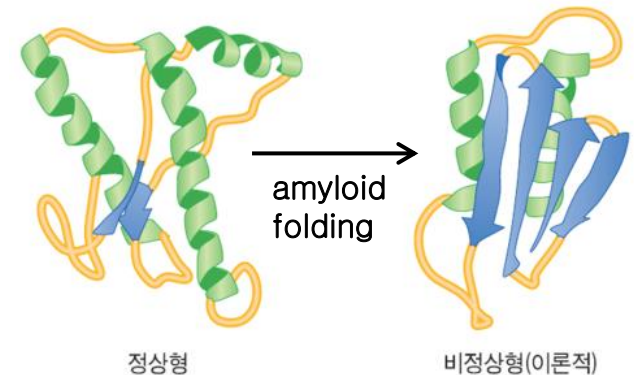
► Basic information...

- Terms in virology

- prion:

- In 1982, discovered by Stanley B. Prusiner
 - infectious agent composed entirely of protein material, called PrP (short for **p**rion **p**rotein)
 - : **p**roteinaceous infectious particle(–on) or **p**rotein and infection
 - PrP^C (cellular or common): a normal protein found on the cell membranes throughout the body
 - PrP^{Sc} (scrapie): abnormal form (isoform) of PrP^C (beta-sheet)/ refolding (amyloid folding) in PrP^C structure / extremely stable to physical (e.g., heat) and chemical (e.g., proteases) agents
 - mechanism for causing disease: accumulation of amyloid fiber in tissue → tissue damage and cell death

그림 10-72 PrP의 구조적 변형.

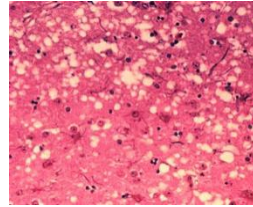


► Basic information...

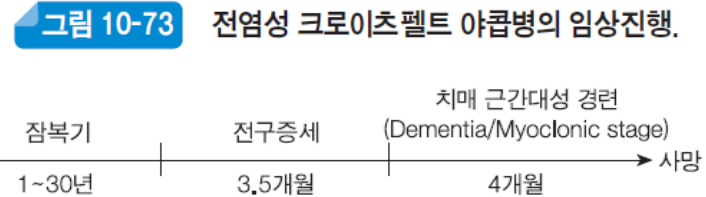
• Terms in virology

– prion:

- neurodegenerative disease
 - by aggregating extracellularly within the CNS to form plaques (known as amyloid) → distrusting the normal tissue structure
 - vacuole (“holes”) formation in neurons → spongy architecture
 - transmissible spongiform encephalopathies:
 - bovin spongiform encephalopathy (mad cow disease)
 - CJD (Creutzfeldt–Jakob Disease; variant CJD) in human
 - symptoms:
 - first symptoms: dementia, personality changes, hallucination, etc.
 - others: anxiety, depression, paranoia, psychosis, etc.
 - physical problems: muscle rigidity, ataxia, seizure, etc.
 - pneumonia due to impaired coughing reflexes
 - fatal within <6 months / 15% surviving for 2 or more yrs (4–5 yrs)
- no vaccines (but effective vaccine in mice) and being untreatable



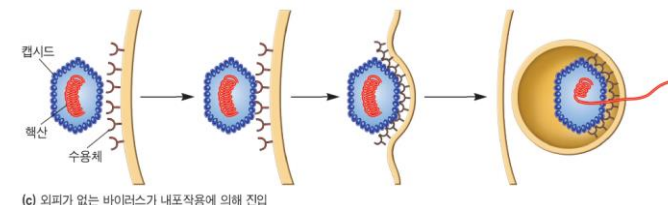
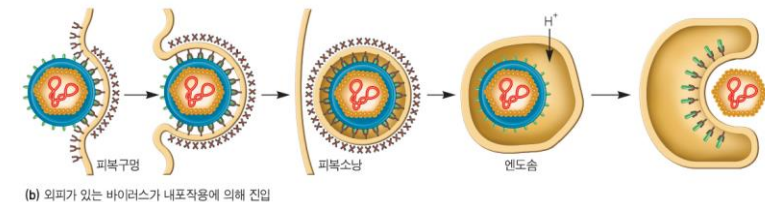
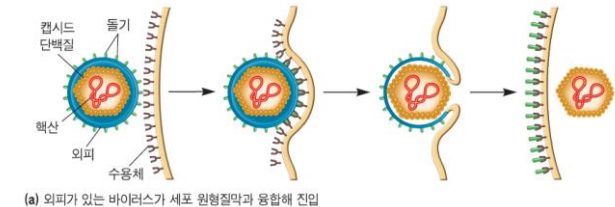
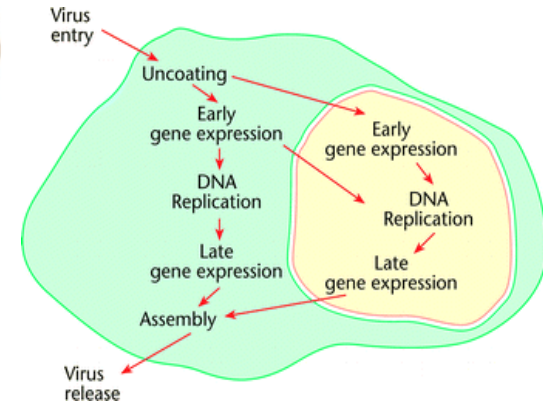
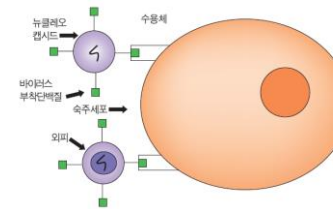
Microscopic “holes”, characteristic in prion-affected tissue sections (“spongy” architecture)



► Basic information

• Virus life cycle

1. **attachment:** interaction of viral surface proteins with molecules on the host cells (called receptors)
2. **penetration and uncoating:** membrane fusion, endocytosis, or genetic injection
3. **genome replication and proteins synthesis**
 - making of copies of itself (genetic materials and own proteins)
 - use of cell's replication machinery or components
4. **assembly and maturation**
5. **release**
 - : release of virus progenies and finding of new hosts



► Basic information

- Virus life cycle

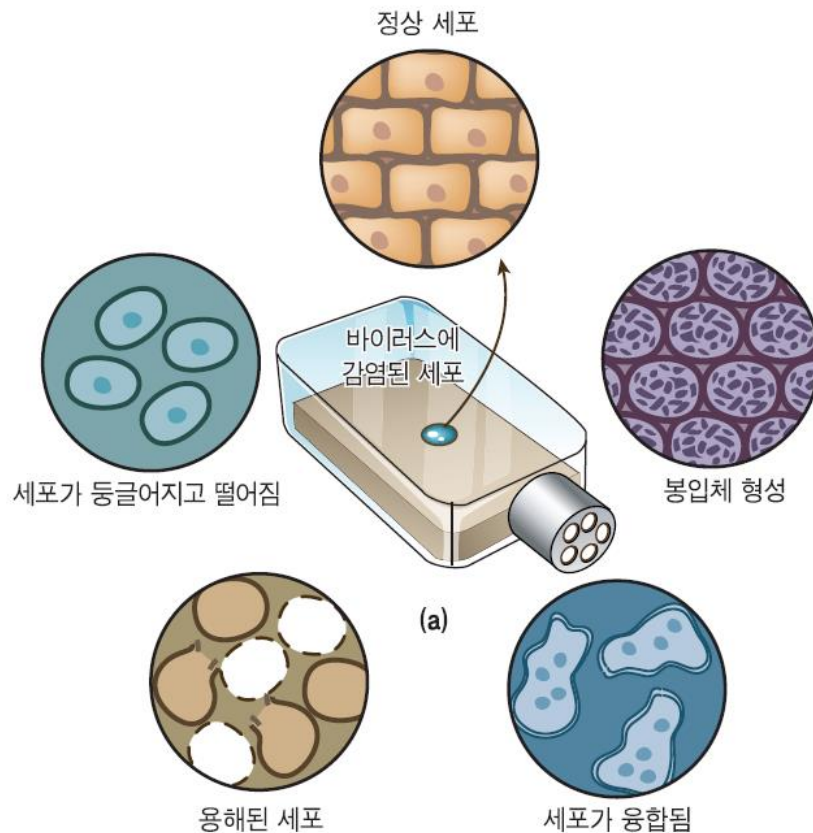
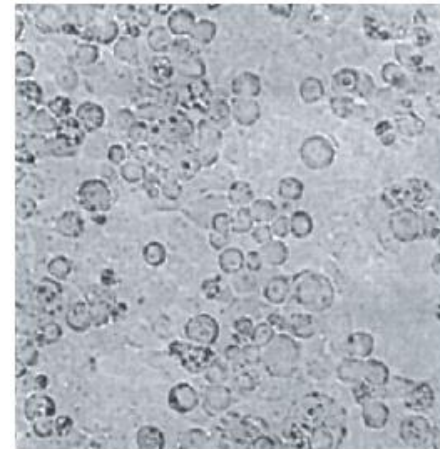
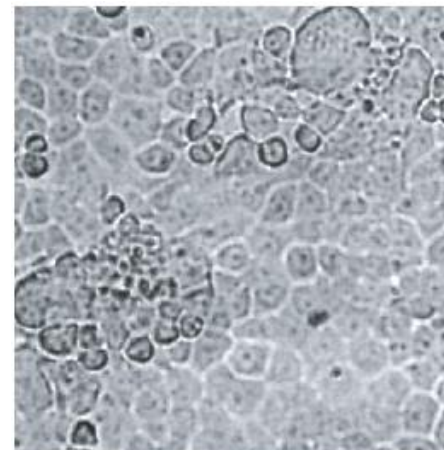


그림 10-2

바이러스 감염으로 인한 세포의 변형.



(b) 바이러스 감염 전 세포



(c) 바이러스 감염 후 세포

► Basic information

- Classification of virus

- nucleic acid types (RNA vs. DNA), presence of envelope (enveloped vs. non-enveloped), structure of viral particles and capsids, etc.
- International Committee on Taxonomy of Viruses (ICTV)
 - : Order (*–virales*) > Family (*–viridae*) > Subfamily (*–virinae*) > Genus (*–virus*) > Species (*–virus*)
- Baltimore (7 groups) (based on viral genome replication)
 - : Class I (dsDNA), II (ssDNA), III (dsRNA), IV (+ssRNA), V (–ssRNA), VI (ssRNA–RT), and VII (dsDNA–RT)

► Basic information

• Classification of virus

- Baltimore (7 groups) (based on viral genome replication)
: Class I (dsDNA), II (ssDNA), III (dsRNA), IV (+ssRNA), V (–ssRNA), VI (ssRNA–RT), and VII (dsDNA–RT)
- All viruses must generate **positive strand mRNAs** from their genomes to produce proteins and replicate themselves.

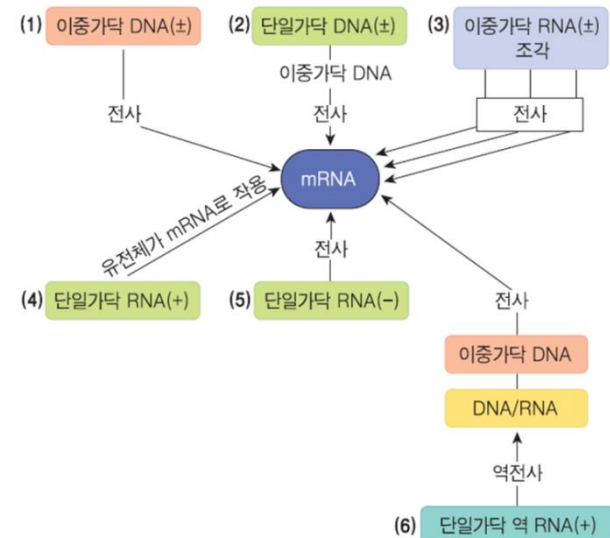
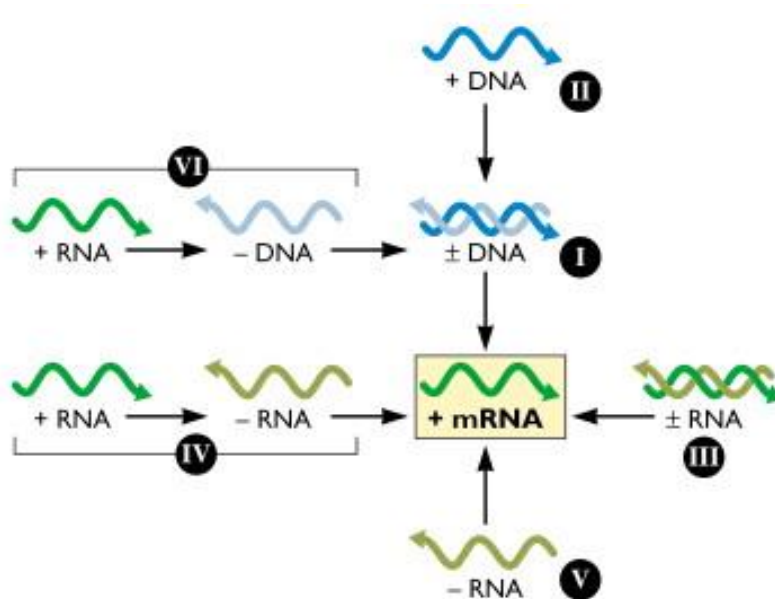


그림 10-11 볼티모어(Baltimore)에 의한 mRNA 생성의 방법.

► Viral groups (classes) and their representative members

- **Group 1:** dsDNA viruses – Poxviridae, Herpesviridae, Adenoviridae, Papillomaviridae, Polyomaviridae, etc.
- **Group 2:** ssDNA viruses – Parvoviridae, etc.
- **Group 3:** dsRNA viruses – Reoviridae, etc.
- **Group 4:** (+) ssRNA viruses – Coronaviridae, Picornaviridae, Caliciviridae, Flaviviridae, Togaviridae, Hepevirus, etc.
- **Group 5:** (–) ssRNA viruses – Paramyxoviridae, Phabdoxviridae, Bunyaviridae, Orthomyxoviridae, etc.
- **Group 6:** (+) ssRNA–RT virus – Retroviridae, etc.
- **Group 7:** dsDNA–RT virus – Hepadnaviridae, etc.

Class 1: dsDNA virus

- ▶ Replication of double-stranded DNA virus
 - DNA-dependent DNA / RNA polymerase

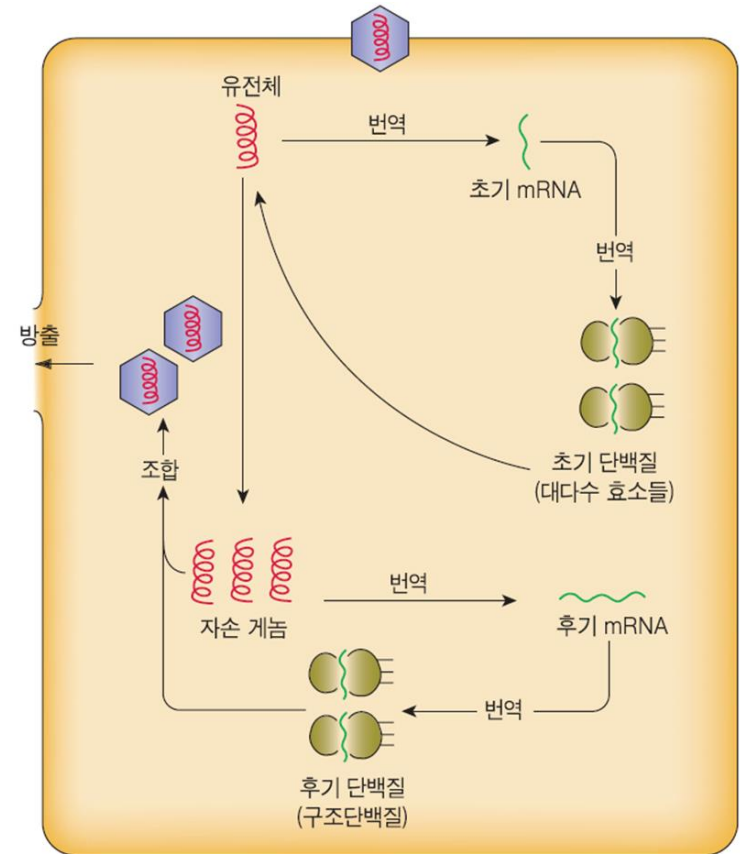
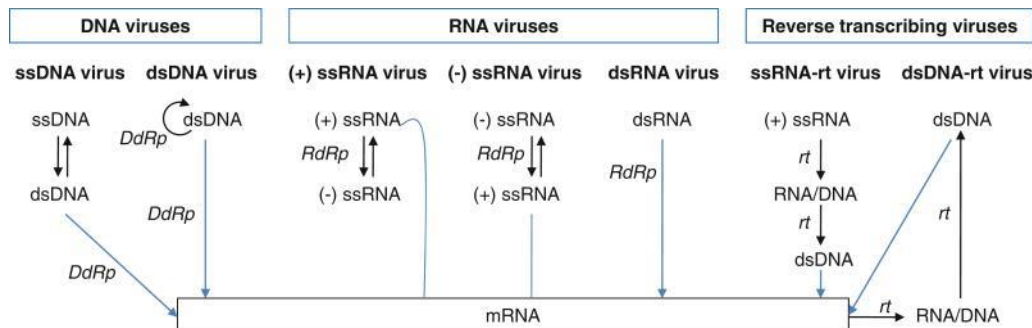
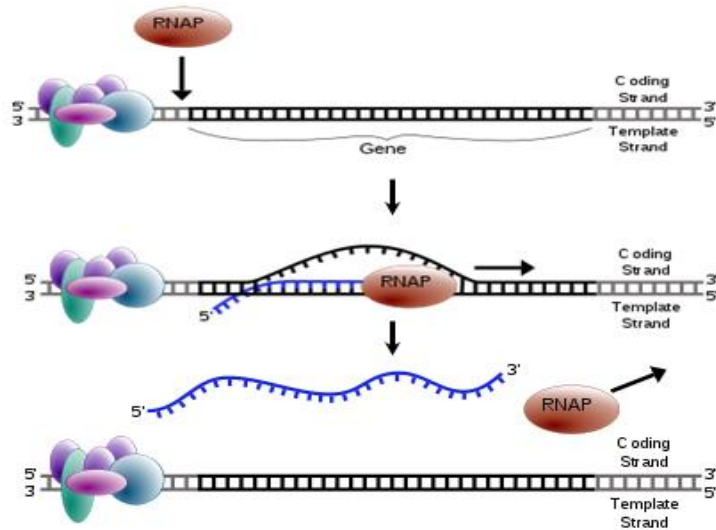


그림 10-12 이중가닥 DNA 바이러스의 증식과정.

► Representative members

- Order Caudovirales
 - Families: Myoviridae, Podoviridae, Siphoviridae
 - Tailed bacteriophages; infect bacteria
- Order Herpesvirales
 - Families: Alloherpesviridae, **Herpesviridae**, Malacoherpesviridae
- Unassigned families
 - Ascoviridae, **Adenoviridae**, Asfarviridae, Baculoviridae, Coccolithoviridae, Corticoviridae, Fuselloviridae, Guttaviridae, Iridoviridae, Lipothrixviridae, Mimiviridae, Nimaviridae, **Papillomaviridae**, Phycodnaviridae, Plasmaviridae, **Polymaviridae**, **Poxviridae**, Rudiviridae, Tectiviridae
- Unassigned genera

Poxviridae

► Family *Poxviridae*

- **Key words:** complex virus (→ physical and genetic autonomy from the host cell; unusual cytoplasmic DNA virus), small pox (variolar virus), vaccinia virus, oncogenicity, etc.
- **Structure:**
 - enveloped; one of the largest known viruses (200–400 nm long)
 - very complex: almost approaching the level of a simple cell
 - unique structure of capsid: apparently lacking any of the symmetry features (being generally shaped like a brick or as an oval form)

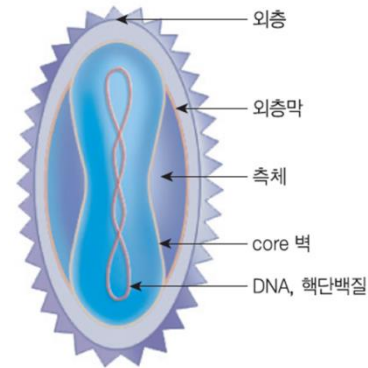
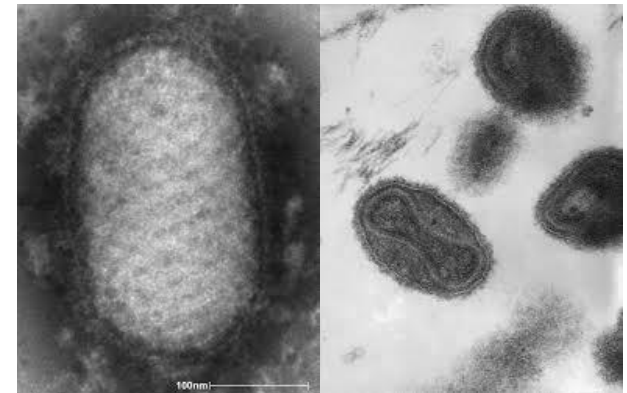
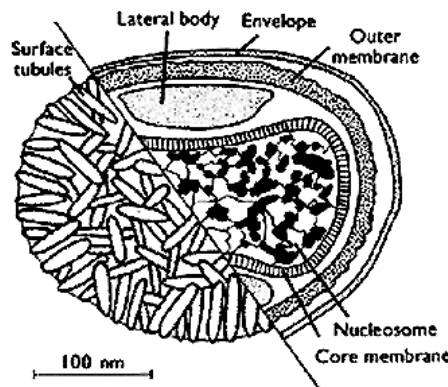
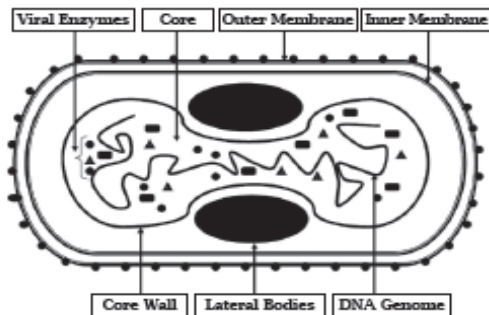


Figure 1: Illustration of the morphology of the *Vaccinia virus*.

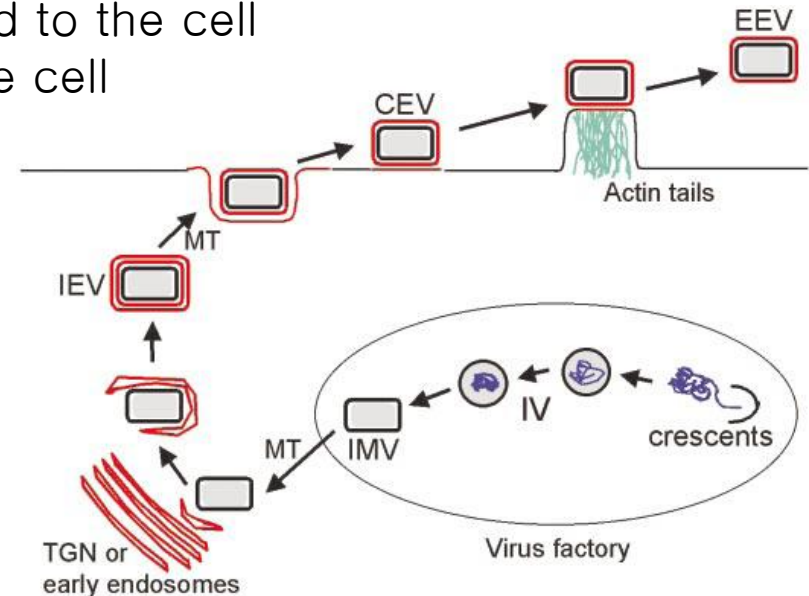
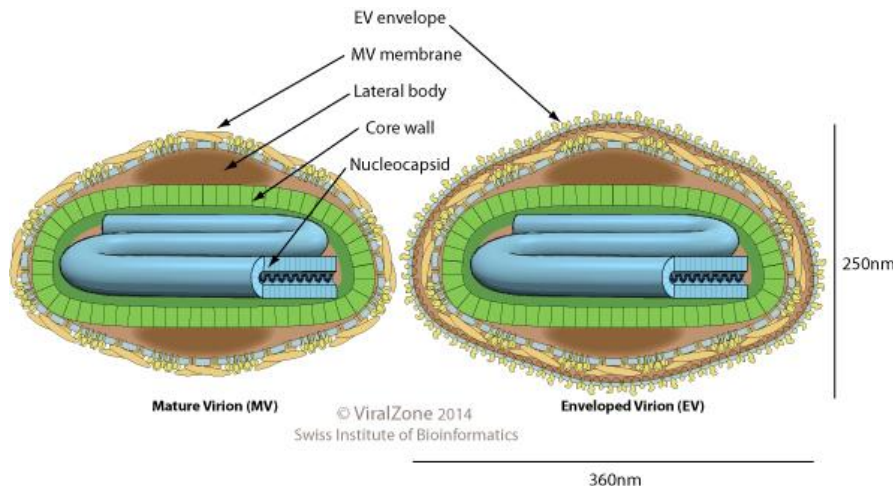


► Family *Poxviridae*

- Structure:

- virion morphogenesis (4 different types of virion):

- IMV: a form of virus being retained in cells until lysis / having an envelope (of which origin is unknown) → transported to the Golgi apparatus
- IEV: an intermediate b/w IMV and CEV / formed by wrapping of IMV with intracellular membranes → transported along cytoskeletal microtubules to cell periphery → fusing with the plasma membrane
- CEV: virus particle remaining attached to the cell
- EEV: virus particle being free from the cell



► Family *Poxviridae*

• Structure:

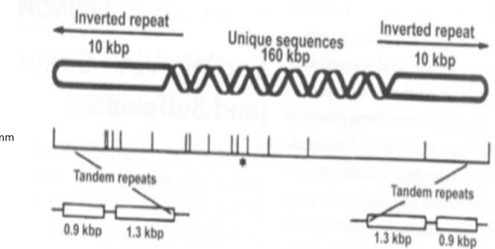
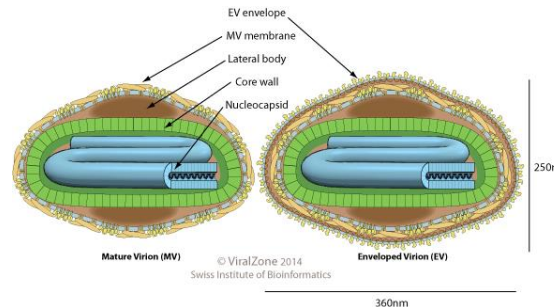
– component of virus particle:

• surface membranes:

- internal membrane: membrane of IMV; unknown origin
- external membrane: envelope of IEV/CEV/EEV; originated from internal cell membrane

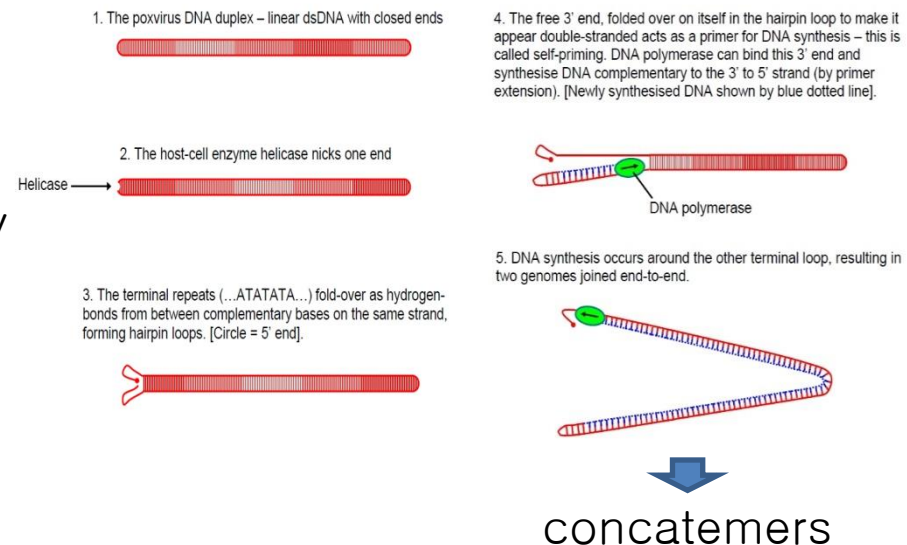
• lateral bodies: lens-shaped and nested b/w the core membrane and the surface membrane / unknown function

• core: tightly compressed nucleoprotein; containing DNA and viral enzymes



• Genome and replication:

- linear, dsDNA (of 130–375 kb) being flanked by inverted terminal repeat (ITR) sequence, which are covalently –closed at their extremities
- encoding viral proteins involved in DNA replication (e.g., DNA pol)
→ replicating in cytoplasm
(unusual cytoplasmic DNA virus)



► Classification of the family *Poxviridae*

Subfamilies	Genera	Members or hosts
<i>Chordopoxvirinae</i>	Orthopoxvirus	Variola, vaccinia, cowpox, monkeypox, camelpox
(vertebrates)	Parapoxvirus	Sealpox, parapox of deer, pseudocowpox
	Avipoxvirus	Canarypox, fowlpox, pigeonpox, turkeypox, penguinpox
	Capripoxvirus	Goatpox, sheeppox
	Leporipoxvirus	Hare fibroma, myoma, rabbit fibroma, squirrel fibroma
	Suipoxvirus	Swinepox
	Molluscipoxvirus	Molluscum contagiosum
	Yatapoxvirus	Tanapox, Yaba monkey tumor
<i>Entomopoxvirinae</i>	Alphaentomopoxvirus	beetles
(insects)	Betaentomopoxvirus	butterflies, moths, grasshoppers, locusts, etc.
	Gammaentomopoxvirus	flies, mosquitoes

► Subfamily *Chordopoxvirinae* (chordo- = chordate: 척삭동물)

1. Genus Orthopoxvirus:

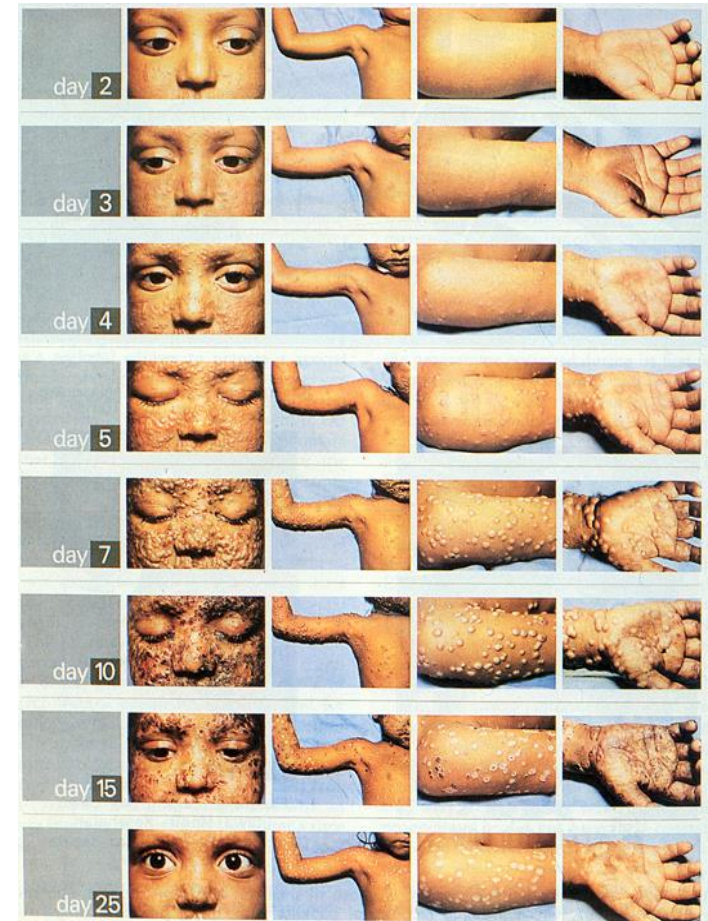
- infects various mammals
: buffalo, camel, cow, monkey, rabbit, mouse (volepox, ectromelia virus – mousepox), and human
- human: variola virus (smallpox; 천연두)
 - etymology: L. *varius* (“spotted”) or *varus* (“pimple”)
 - name: smallpox vs. great pox (syphilis)
 - history:
 - earliest evidence: Egyptian mummies (Ramses V; 1157 BC)
 - victim in the 20th century alone: 300 million people
 - total fatalities: probably near 1 billion
 - last outbreak: occurring in an outbreak of two cases (one of which was fetal) in Birmingham, UK in 1978
 - highly contagious / human-to-human transmission (via respiratory tracts)
 - spread: – 1st viremia: respiratory tract → blood
– 2nd viremia: blood → multiple organs → blood → skin
 - two strains:
 - major strain: severe / mortality rate of 10–30% (~20%)
 - minor strain: mild / mortality rate of 1–2%



► Subfamily *Chordopoxvirinae*

1. Genus Orthopoxvirus:

- human: variola virus (smallpox; 천연두)
 - symptoms:
 - ordinary: similar to influenza and common cold (e.g. fever, muscle pain, malaise, headache, prostration), nausea, vomiting, backache → small reddish spots on mucous membranes of the mouth, tongue, and throat → maculopapular skin rash (반구진 or 반점성 발진) → pock mark (반흔)
 - modified: less severe than in the ordinary type
 - malignant: flat smallpox / more severe than in the ordinary type (e.g. mortality of ~80%)
 - hemorrhagic: the most severe form / extensive bleeding into the skin / usually fatal within 1 week



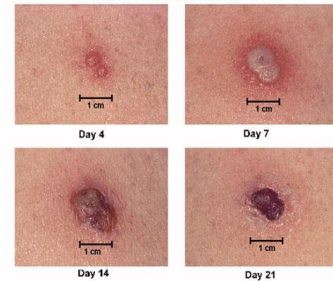
► Subfamily *Chordopoxvirinae*

1. Genus Orthopoxvirus:

- human: variola virus (smallpox; 천연두)
 - control: vaccination by live-attenuated vaccinia virus
 - history of vaccination and eradication:
 - Edward Jenner (1796):
 - cowpox infections: nearly universal in milkmaids / causing a mild infection in humans
 - observing milkmaids rarely contracting smallpox
 - inoculating people by scarification with cowpox crusts (termed variolation; 우두접종법)
 - global eradication program: proposed by the Soviets in the 1950s
 - intensified smallpox eradication program: begun by the WHO
 - being declared eradicated in 1979
 - effective vaccine: only one serotype
 - long-term immunity



Primary Vaccination Site Reaction



► Subfamily *Chordopoxvirinae*

1. Genus Orthopoxvirus:

- human: variola virus (smallpox; 천연두)
 - some nations being supposed to have smallpox virus as a biological weapon:
 - USA and Russia
 - some other nations: North Korea, Syria, Iran, Israel
 - current smallpox vaccine supply:
 - US government has 15.4 million doses of Dryvax vaccine
 - Additional 85 million doses (Aventis Pasteur) held for emergency use

► Subfamily Chordopoxvirinae

2. Genus Parapoxvirus

- infects various mammals
- having a unique spiral coat
- Orf virus: sheep, goat, human
- Pseudocowpox virus: cattle, human
- Bovine papular stomatitis virus: cattle, human
- Seal parapoxvirus: seal, human
- Ausdyk virus: camel



3. Genus Avipoxvirus (avi- = avian: 새)

- infects birds (fowl, pigeon, quail, sparrow, etc.)
- withstand extreme dryness



► Subfamily *Chordopoxvirinae*

4. Genus Capripoxvirus (capri- = goat: 염소)
 - Sheeppox virus
 - Goatpox virus
 - Lumpyskin disease virus: cattle, buffalo
5. Genus Suipoxvirus (sui- = swine: 돼지)
 - Swinepox virus
6. Genus Leporipoxvirus (lepori- = leporid: 토끼)
 - Myxoma virus: rabbit
7. Genus **Molluscipoxvirus** (mollusci- = molluscum: 연속종)
 - Molluscum contagiosum virus: human
8. Genus **Yatapoxvirus**
 - **Y**abapox virus (Yaba monkey tumor virus): monkey, human
 - **T**anapox virus: monkey

Table 15.1 Poxviruses pathogenic for humans

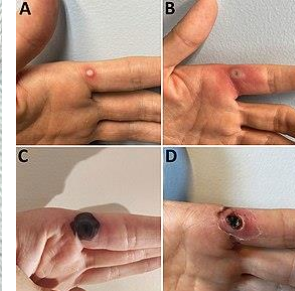
Genus	Virus	Reservoir host	Animals naturally infected	Geographical distributed	Comment
<i>Orthopoxvirus</i>	Smallpox	Humans	—	Formerly worldwide	Last endemic case 1977 Eradication confirmed 1979
	Monkeypox	Squirrels	Squirrels Monkeys	West and Central Africa	Rare zoonosis. Overall mortality 10%. Limited case-to-case spread
	Cowpox	Rodents	Cats Cattle	Europe, West CIS (USSR)	Rare zoonosis. Contact with cattle unusual
	Vaccinia (buffalopox)	Buffalo? Rodents?	Buffalo	India	Variant of vaccinia. Established in nature
<i>Parapoxvirus</i>	Orf	Sheep	Sheep	Worldwide	Common trivial zoonoses. Occupational hazards
	Pseudocowpox	Goats Cattle	Goats Cattle		
<i>Molluscipoxvirus</i>	Molluscum	Humans	—	Worldwide	Trivial infection. Often sexually transmitted
<i>Yatapoxvirus</i>	Tanapox	Monkeys	Monkeys	Kenya, Zaire (Congo Republic)	Rare trivial zoonosis

► Subfamily *Chordopoxvirinae*

1. Genus Orthopoxvirus:

– monkeypox (currently, Mpox)

- infectious disease (rare zoonosis) caused by monkeypox virus
 - animal contact
 - infected person contact: *face-to-face* (talking or breathing; respiratory droplets or short-range aerosols from prolonged closed contact), *skin-to-skin* (touching or vaginal/anal sex), *mouth-to-mouth* (kissing), *mouth-to-skin contact* (oral sex or kissing the skin), etc.
 - materials; contaminated sheets, clothes or needles
 - mother-to-fetus: pregnant persons, who may pass the virus on their unborn baby
- symptoms: (incubation period) 5 ~ 21 days / **(flu-like)** fever, sore throat, cough, headache, muscle ache, back pain → swollen lymph nodes → **skin rash** (palms of hands and soles of feet / face, mouth and throat / groin and genital areas / anus)
- fatality rate in humans: 1–10% of cases
 - west Africa: 1–2%
 - Democratic Republic of Congo: 10%



► Subfamily *Chordopoxvirinae*

1. Genus Orthopoxvirus:

– monkeypox (currently, Mpox)

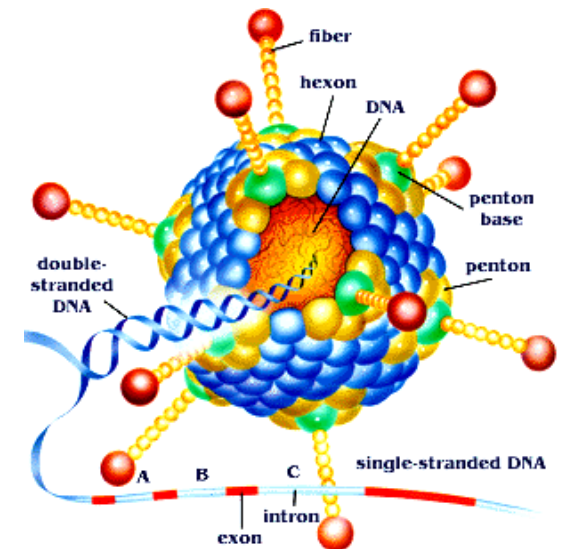
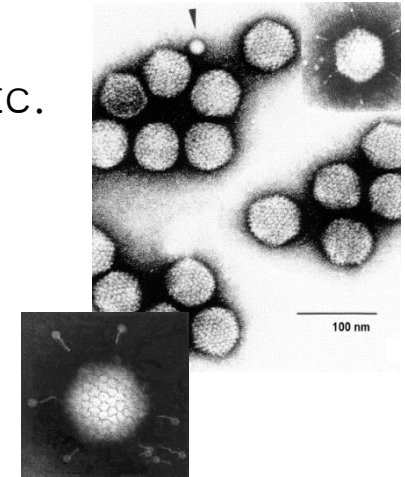
- first identified in 1958 among laboratory monkeys in Copenhagen, Denmark
- first human cases in 1970: a 9-month-old boy in Congo (DRC)
 - over 300 human cases (mainly, by contact with animals)
- 72 cases of an US outbreak in 2003
 - Wisconsin, Illinois, Indiana, Kansas, Missouri, Ohio, etc.
 - introduced from Africa to US pet supplier
 - : Gambian giant rat, Prairie dogs
- 2022–2023 mpox outbreak
 - initially identified in the UK in May 2022
 - widespread community transmission outside of Africa
 - WHO, declaring a Public Health Emergency of International Concern (PHEIC) (2022.07~2023.05)
 - (as of August 27, 2023) subsequent 89,596 cases confirmed (157 death) in 114 countries



Adenoviridae

► Family Adenoviridae

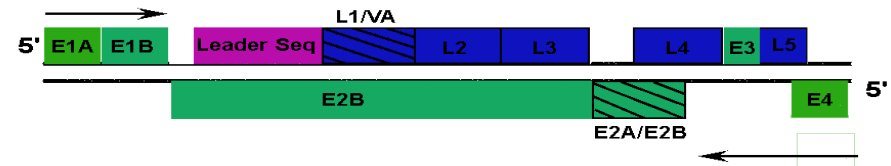
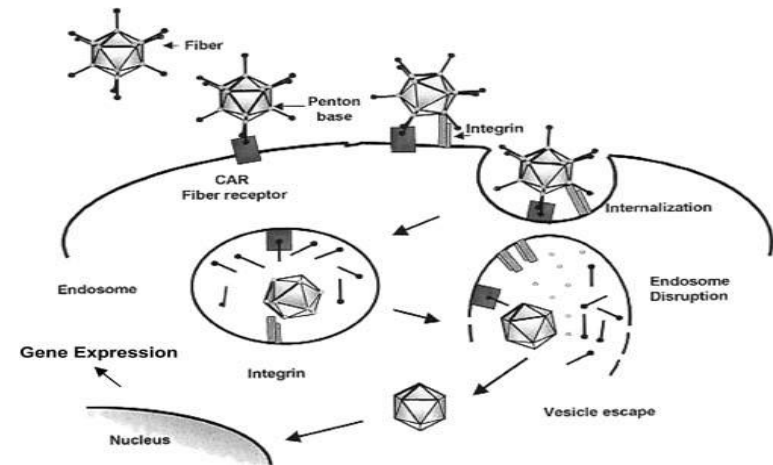
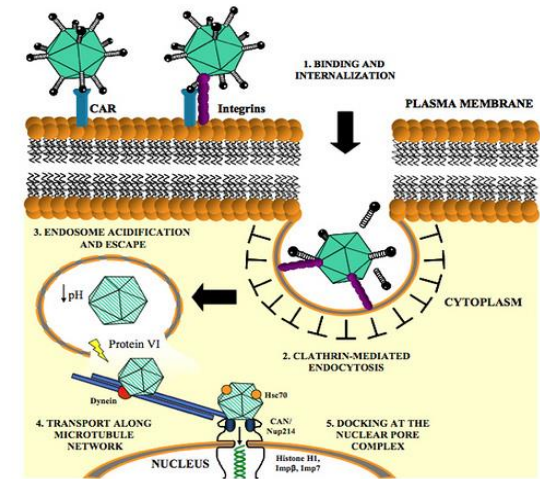
- **Key words:** common cold, oncogenicity, role as vectors, etc.
- **Etymology:**
 - derived from their initial isolation from human adenoids
 - Adeno- = “gland”
- **Structure:**
 - non-enveloped: the largest non-enveloped virus
 - icosahedral nucleocapsid with 252 capsomers
 - 12 pentons at vertices + 240 hexons
 - each penton having a fiber
 - medium-sized (90–100 nm), but the largest non-enveloped virus
- **Properties:**
 - stable in the environment
 - relatively resistant to disinfection (e.g., alcohol, detergent, etc.)
 - stable in gastrointestinal tract: withstanding low pH, bile acid and proteolytic enzymes



► Family Adenoviridae

• Genome and replication:

- ds linear DNA / replicating in nucleus
- receptor-mediated endocytosis
 - fiber vs. receptor and co-receptor
 - coxsackievirus-adenovirus receptor (CAR) (+CD46, MHC, sialic acid)
 - co-receptor: αv integrin
- gene expression
 - early phase: E1A, E1B, E2, E3, E4
 - delayed early: IVa2, IX
 - late phase: L1–L5
- **oncogenic proteins: E1A & B**
(interacting with tumor suppressor protein p53; in vitro only)
- **immune evasion-related protein**
: E3 (inhibiting MHC I Ag expression)



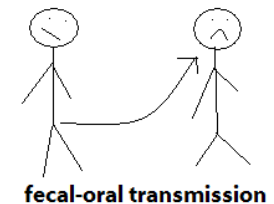
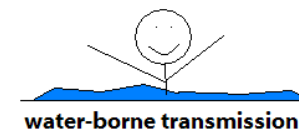
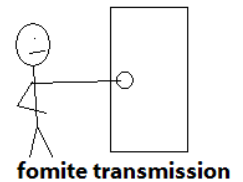
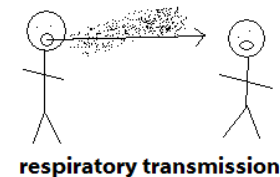
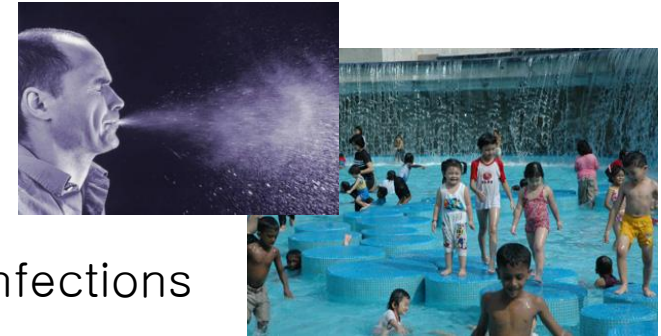
► Family Adenoviridae

- Clinical signs:

- responsible for 5–10% of upper respiratory infections
- mild to severe (life-threatening) symptoms
 - mild respiratory infections: fever, tonsillitis, eye (conjunctivitis) or ear infection
 - severe multiorgan disease: gastroenteritis, hemorrhagic cystitis / bronchiolitis, pneumonia, meningitis, encephalitis, symptom similar to whooping cough (cf. *Bordetella*)

- Transmission:

- primarily, respiratory droplets
 - left on objects ('fomite'): towels, cups, etc.
- fecal-oral route → stools
- poorly chlorinated water
- etc.



► Family Adenoviridae

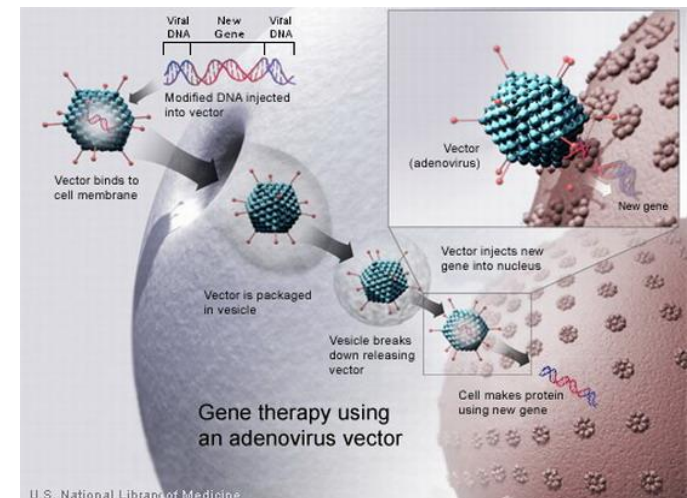
• Control:

- previously, vaccines against two serotypes of human adenovirus being used in US military, but currently, no vaccines available
- no antiviral drugs
- treatments by doctors: being mainly directed at the symptoms or preventing secondary bacterial infections
- the best way to control: good handwashing



• Role as vectors

- recombinant DNA vaccines
 - gene therapy
 - E3: not essential for replication in cell culture
- Insertion site for foreign DNA



► Family Adenoviridae

- **Classification:** two genera

1. Mastadenovirus: infect mammalian species

- Human adenovirus: 57 serotypes (in 6 subgroups (A–F))
- Simian adenovirus: 12 serotypes
- Bovine adenovirus: 10 serotypes
- etc.

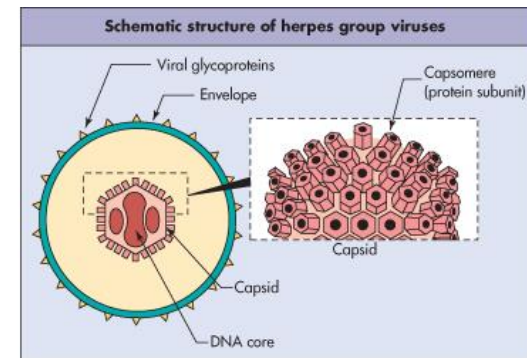
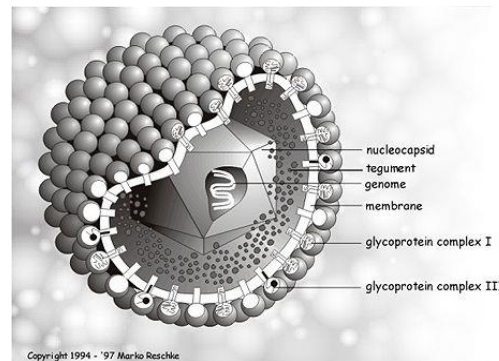
2. Aviadenovirus: infect avian species (birds)

- Fowl adenovirus: 12 serotypes
- Turkey adenovirus: 3 serotypes
- Goose adenovirus: 3 serotypes
- etc.

Herpesviridae

► Family *Herpesviridae*

- **Key words:** tegument, latent infection, viral DNA polymerase, viral thymine kinase, HSV-1 (cold sore) and -2 (genital herpes), VZV (chickenpox and shingles), HCMV, EBV, KSHV, oncogenicity, etc.
- **Structure:**
 - enveloped → sensitive to acids, detergents, desiccation, ether, etc.
 - icosahedral capsid: ~100 nm in diameter / 162 capsomers
 - all herpesviruses being identical: not be distinguished under EM
 - tegument (viral matrix):
 - a layer of viral proteins b/w the envelop and the capsid
 - tegument proteins: DNA replication or immune evasion

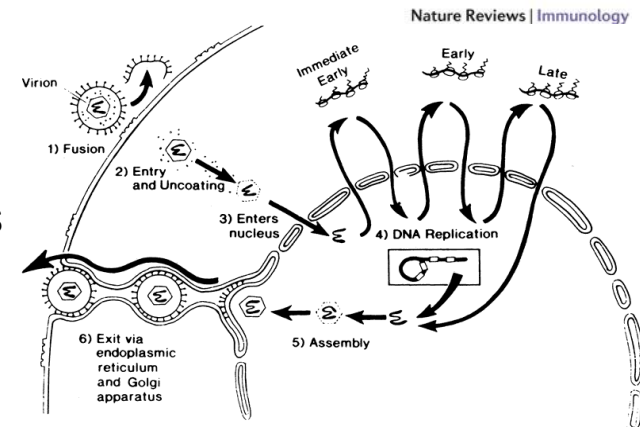
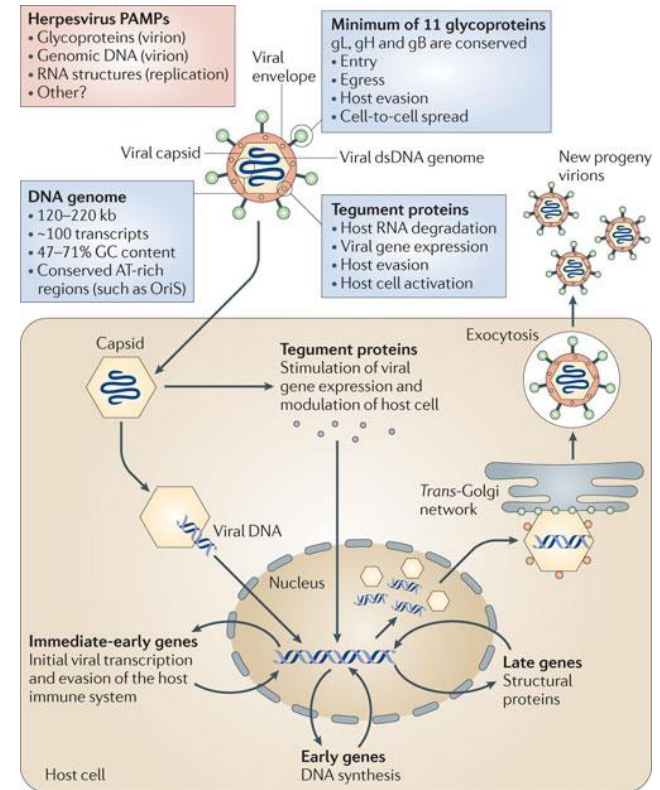
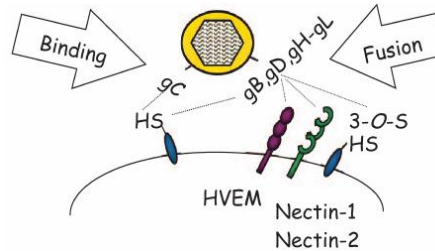


B

► Family *Herpesviridae*

• Genome and replication:

- linear, dsDNA: 124–230 kb in length
- replication in the nucleus (e.g., HSV):
 - binding of multiple viral env. glycoproteins with multiple cell surface receptors
 - fusion of the viral env. with cell memb.
 - release of capsid and tegument proteins (e.g., VP16) into the cytoplasm
 - transport of the capsid to the nuclear pore
 - release of viral DNA into the nucleus
 - circularization of viral DNA
 - tp of viral DNA: by host RNA pol II
 - α gene tp: stimulated by the VP16
 - β gene tp: transactivated by the α proteins
 - γ gene tp: stimulated by viral DNA synthesis
 - structural proteins (e.g., capsid)
 - DNA replication: stimulated by β proteins
 - assembly and budding



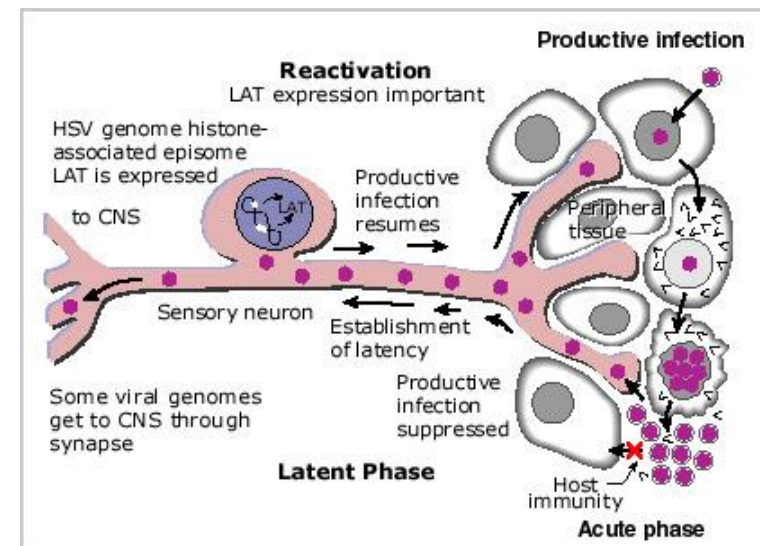
► Family *Herpesviridae*

• Characteristics:

- fragile virion: not survive well outside of the host body
→ transmission: occurring by close or direct contact
- targets: epithelial cells of skin and mucosa, leukocytes and neurons
- latent infection:
 - neurotropism: nerve cell ganglia nearest to the original site of infection
 - latency-associated transcript (LAT): a length of RNA accumulating in cells hosting latent infections / interfering with the normal activities of the infected host cell
 - lifelong persistence
 - etymology: herpes- = G. herpein (meaning “to creep”)

• Classification: 3 subfamilies

- Alphaherpesvirinae: HSV-1, HSV-2, VZV
- Betaherpesvirinae: CMV, Roseolovirus
- Gammaherpesvirinae: EBV, KSHV



► Family *Herpesviridae*

- Subfamily *Alphaherpesvirinae*

- called “herpes simplex virus group”
- two genera: Simplexvirus & Varicellovirus



1. Genus Simplexvirus

- herpes simplex virus 1 (HSV-1) or human herpesvirus (HHV)-1

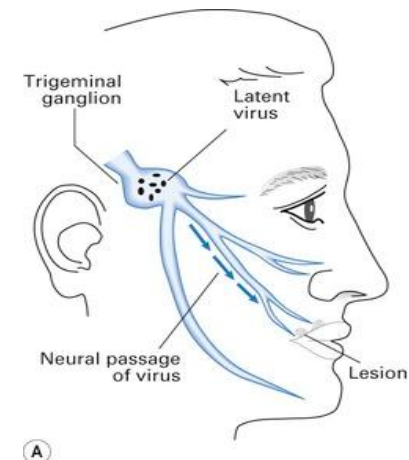
- symptoms:

- mostly oral herpes (herpes labialis or cold sore; 구순포진)
: watery blisters in the skin or mucous memb. of the mouth or lips
- keratoconjunctivitis (각막결막염), encephalitis (뇌염), etc.
- may be related to Alzheimer's disease

- major spread: close contact (with saliva; e.g. sharing drinks and kiss)

- primary infection: mucoepithelial cells

- latent (persistent) infection (neurotrophic): nerves at the site of primary infection → cell body of the neuron → trigeminal ganglion (3차 신경절)

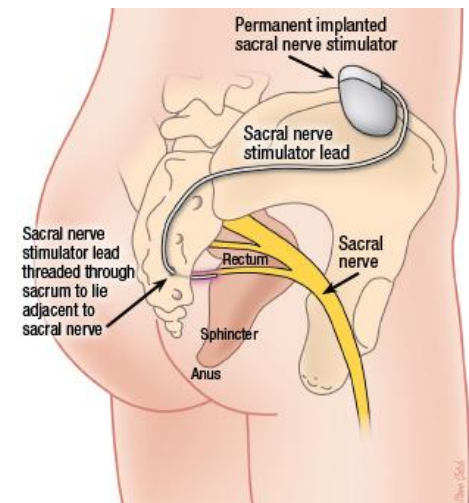
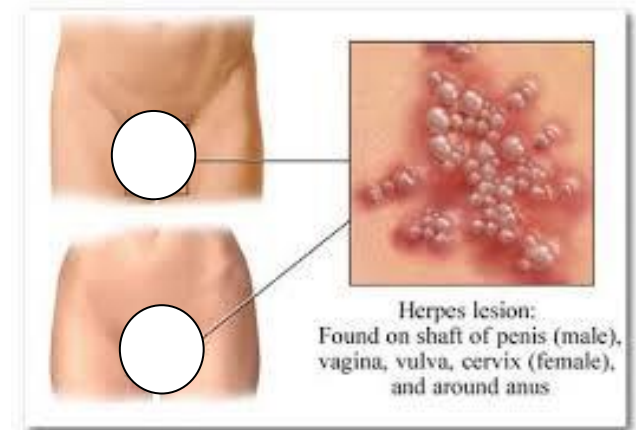


► Family *Herpesviridae*

- Subfamily *Alphaherpesvirinae*

1. Genus Simplexvirus

- **herpes simplex virus 2 (HSV -2) or HHV-2**
 - symptoms: mostly genital herpes
 - watery blisters in skin or mucous membranes of genitals
 - major spread: close contact (e.g. sexual practices)
 - primary infection target: mucoepithelial cells
 - latent (persistent) infection target: sacral ganglion (엉치 신경절)
- both HSV-1 and HSV-2:
 - infecting either oral or genital sites
 - HSV-1: more likely to reactivate from oral sites
 - HSV-2: more likely to reactivate from genital sites



► Family *Herpesviridae*

- Subfamily *Alphaherpesvirinae*

- 1. Genus Simplexvirus

- **diagnosis:**

- direct detection
 - electron microscopy: rapid, but not distinguish b/w species of virus
 - cytopathology: observing multinucleated giant cells from skin scrapings / not distinguish b/w HSV and VZV
 - immunofluorescence: viral proteins using their antibodies / not distinguish b/w HSV-1 and -2, but do b/w HSV and VZV
- virus isolation:
 - cultivation: both are among the easiest viruses to cultivate / taking 1–5 days
 - restriction digestion of viral DNA : discriminating HSV-1 from HSV-2
- PCR: used for systemic or encephalitic disease
- serology: appearance of IgG in 4–7 days / not discriminate HSV-1 vs.-2

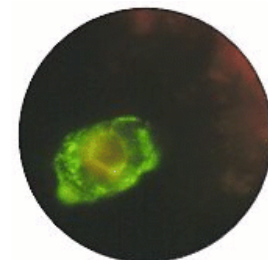
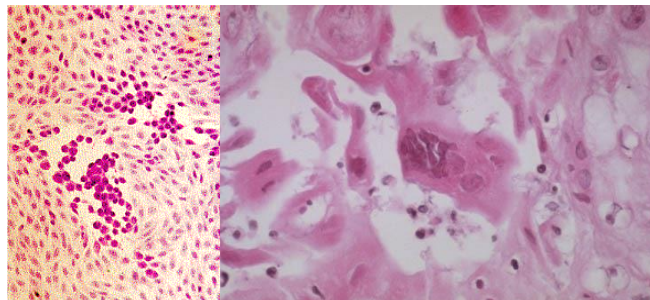


Fig. 3, HSV-infected epithelial cell from skin lesion (DFA)

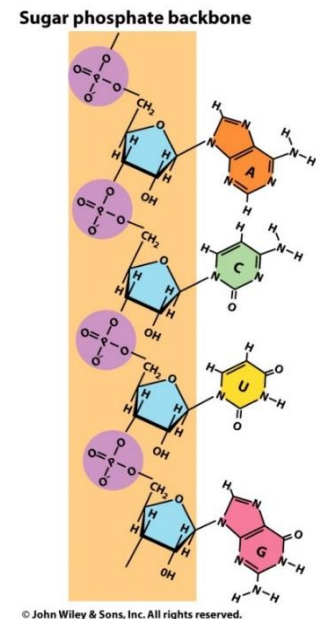
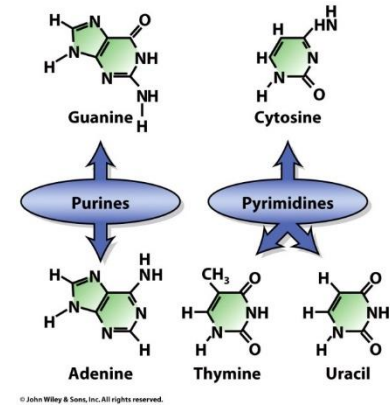
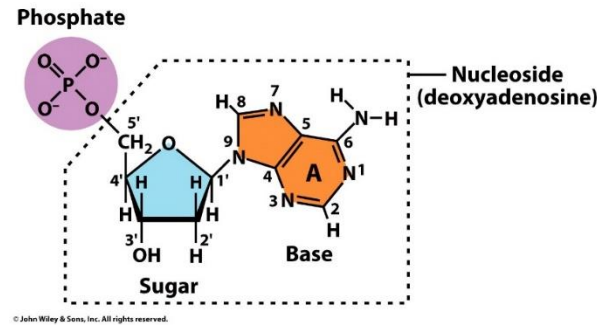
► Family *Herpesviridae*

- Subfamily *Alphaherpesvirinae*

1. Genus Simplexvirus

- control methods:

- no vaccines, but antiviral drugs available
- nucleoside analogues targeting for viral enzymes (DNA polymerase)
- nucleotide: basic building block of DNA / sugar + base + phosphate
- nucleoside: sugar + base
 - adenine / adenosine / deoxyadenosine
 - guanine / guanosine / deoxyguanosine
 - thymine / 5-methyluridine / thymidine
 - uracil / uridine / deoxyuridine
 - cytosine / cytidine / deoxycytidine



► Family *Herpesviridae*

- Subfamily *Alphaherpesvirinae*

1. Genus Simplexvirus

- control methods:

- nucleoside analogues targeting for viral enzymes (DNA polymerase)

- acyclovir (ACV) and its derivatives (e.g. valacyclovir): a guanosine analogue / active for HSV-1 and -2 (+VZV, CMV, EBV, etc.) / various formulations (intravenous, oral, cream, ophthalmic ointment, etc.) for treatment and prophylaxis

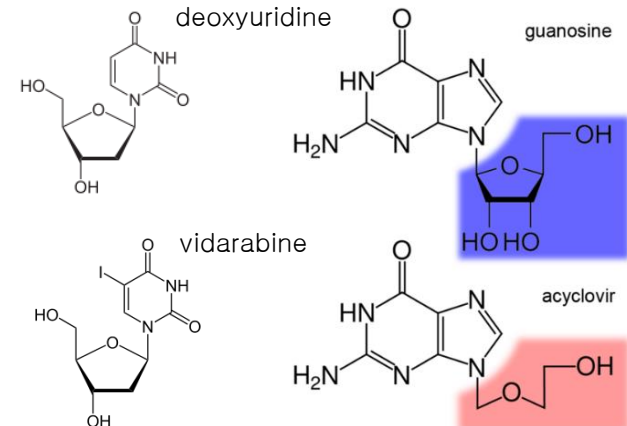
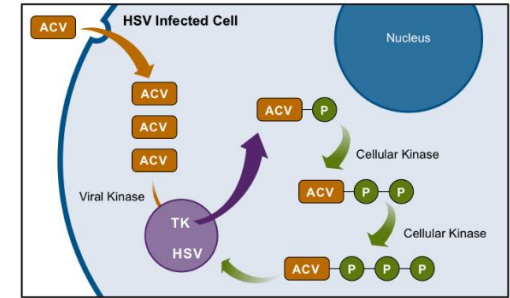
- penciclovir and its derivatives (e.g. famciclovir): a guanosine analogue
 - penciclovir: low toxicity and good selectivity, but absorbed poorly when given orally

- famciclovir: improved oral bioavailability

- vidarabine: an adenosine analogue

- idoxuridine/trifluridine: deoxyuridine analogues / active for HSV-1, but not for HSV-2 and VZV

- mechanism of prodrugs: activation of the drugs by phosphorylation by viral thymine kinase
→ binding with viral DNA polymerase



► Family *Herpesviridae*

- Subfamily *Alphaherpesvirinae*

2. Genus *Varicellovirus*

- varicella-zoster virus (Human herpesvirus 3 or HHV-3)

- diseases:

- chickenpox (수두): also called (primary) varicella

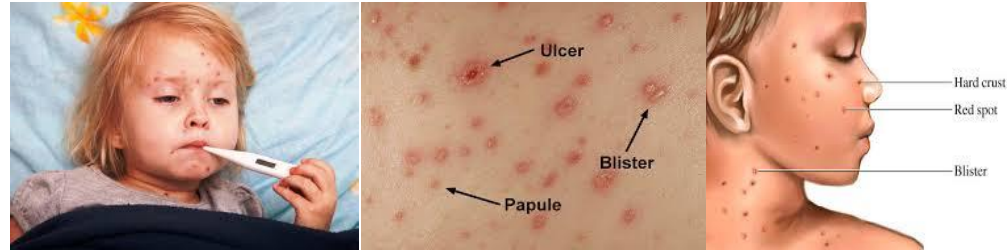
- symptoms: fever, vesicular rash, lymphadenopathy, etc.

- an endemic disease:

- one of the classic diseases of childhood (4–10 years old age group)
- increased severity in adults and immunocompromised patients

- complications: being rare

- common case: secondary bacterial infection of the vesicles
- rare cases: viral pneumonia, encephalitis, and haemorrhagic chickenpox



► Family *Herpesviridae*

- Subfamily *Alphaherpesvirinae*

2. Genus *Varicellovirus*

- varicella-zoster virus (HHV-3)

- diseases:

- **shingles** (대상포진): also called (herpes) zoster

- recurrent infection of chickenpox (10–20% of individuals)
 - reactivation of the latent virus
 - occurring sporadically and evenly throughout the year
 - may occur at any age, but mainly do more than 50 years of age
- symptoms: characteristic eruption of vesicles in the dermatome, postherpetic neuralgia (intensive pain lasting for months), etc.
- severity:
 - great problems caused by affecting the eye and face
 - in immunocompromised patients: occurring earlier in life, multiple attacks, and complications (rare, but including encephalitis and disseminated herpes zoster)



► Family *Herpesviridae*

- Subfamily *Alphaherpesvirinae*

2. Genus *Varicellovirus*

- varicella-zoster virus (HHV-3)

- diseases:

- **congenital VZV infection**

- primary infection during pregnancy:
 - 90% of pregnant women: being already immunity
 - being rare, but a greater risk of severe disease (e.g, pneumonia)
- congenital varicella syndrome:
 - first 20 weeks of pregnancy
 - up to 3% chance to transmission to the fetus
 - scarring of skin, hypoplasia of limbs, CNS and eye defects, death in infant normal, etc.



► Family *Herpesviridae*

- Subfamily *Alphaherpesvirinae*

2. Genus *Varicellovirus*

- varicella–zoster virus (Human herpesvirus 3 or HHV–3)
 - major spread: gain entry via respiratory tract and spreads to the lymphoid system (incubation time: 14–21 days)
 - primary target: mucoepithelial
 - site of latency: neuron (cerebral or posterior root ganglia)
 - diagnosis: direct detection, virus isolation, and serology
 - control: antiviral drugs (e.g. acyclovir, famciclovir, valacyclovir, etc.), vaccine (live–attenuated; one antigenic serotype only), INF, VZV immunoglobulin, etc.
- Bovine herpesvirus 1 and 5, Caprine herpesvirus 1, Porcine herpesvirus 1, Equine herpesvirus 1, 3 and 4, etc.

► Family *Herpesviridae*

- Subfamily *Betaherpesvirinae*

1. Human cytomegalovirus (HCMV) or human herpesvirus 5 (HHV-5)

- one of the most successful human pathogens: able to be transmitted vertically and horizontally usually with little effect on the host → (once infected) carrying it for life
 - infection rate:
 - 40% of adolescents / ~70% of the population in developed countries (with a high standard of hygiene)
 - >90% of the population in developing countries
 - transmission:
 - close contact with the saliva and urine (sexual transmission, blood and blood products, transplanted organ, etc.)
 - congenitally (in utero), perinatally or postnatally
- perinatal infection: acquired mainly through infected genital secretions / 10 times more common than congenital infection

► Family *Herpesviridae*

- Subfamily *Betaherpesvirinae*

1. Human cytomegalovirus (HCMV) or human herpesvirus 5 (HHV-5)

- symptoms: from asymptomatic (healthy people) to life-threatening (immunocompromised people)
 - perinatal infection: usually asymptomatic
 - postnatal infection: usually asymptomatic, but may cause syndrome of infectious mononucleosis
 - infectious mononucleosis: fever, lymphadenopathy, splenomegaly, etc.
 - severe CMV disease in immunocompromised patients (e.g., transplant recipients and AIDS)
 - : pneumonitis, retinitis, colitis, encephalopathy, etc.
 - congenital infection: may cause cytomegalic inclusion diseases

► Family *Herpesviridae*

- Subfamily *Betaherpesvirinae*

1. Human cytomegalovirus (HCMV) or human herpesvirus 5 (HHV-5)

- symptoms:

- congenital infection:

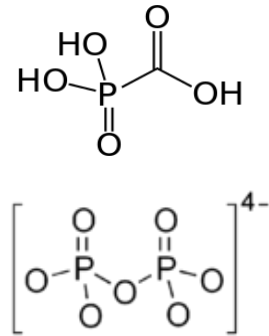
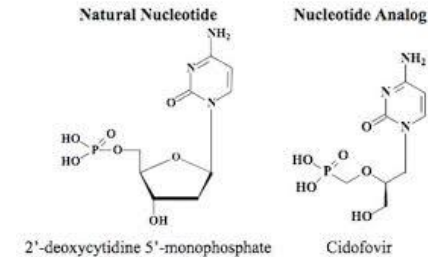
- 40% of chance of transmission to the fetus following a primary or recurrent infection
 - cytomegalic inclusion disease: CNS abnormalities (microcephaly, mental retardation, spasticity, epilepsy, periventricular calcification), eye (choroidoretinitis, optic atrophy), ear (sensorineural deafness), liver (hepatosplenomegaly, jaundice), lung (pneumonitis), heart (myocarditis), thrombocytopenic purpura, haemolytic anaemia, late sequelae in individuals asymptomatic at birth (hearing defects, reduced intelligence)

► Family *Herpesviridae*

- Subfamily *Betaherpesvirinae*

1. Human cytomegalovirus (HCMV) or HHV-5

- primary targets: monocytes, lymphocytes, epithelial cells
- sites of latency: monocytes, lymphocytes
- control:
 - perinatal and postnatal infection: usually not necessary to treat
 - congenital infections: not usually possible to detect / offered the choice of an abortion
 - immunocompromised patients: antiviral drugs (e.g. ganciclovir, valganciclovir, cidofovir, foscarnet, etc.)
 - cidofovir: active for CMV (primarily), acyclovir-resistant HSV, smallpox, VZV (in vitro only), BK virus, etc. / inhibiting viral DNA polymerases by its metabolite (cidofovir diphosphate) / only available as an intravenous formulation
 - foscarnet: active for CMV, HSV, etc. / a pyrophosphate analog – inhibiting viral DNA polymerase by binding to pyrophosphate-binding site (non-nucleoside DNA polymerase inhibitor)



► Family *Herpesviridae*

- Subfamily *Betaherpesvirinae*

2. Muromegalovirus: murin herpesvirus 1 (MuHV-1) and 2 (MuHV-2)

- MuHV-1: mouse cytomegalovirus (MCMV)
- MuHV-1: rat cytomegalovirus (RCMV)

3. Roseolovirus:

- Human herpesvirus 6 & 7
- causing roseola (infantum) (소아 장미진)
 - a childhood disease (6-month- to 2-year-old)
 - (often symptomatic) fever, diarrhea, and exanthem subitum rash
 - (although rare) seizure and encephalitis
- reactivation
 - occurring in locations throughout the body, including the brain, lungs, heart, kidney and gastrointestinal tract
 - (some cases) cognitive dysfunction, permanent disability and death



► Family *Herpesviridae*

- Subfamily *Gammaherpesvirinae*

- Lymphotropic herpesviruses
- Four genera: Lymphocryptovirus, Rhadinovirus, Macavirus, Percavirus

1. Lymphocryptovirus

- **Epstein-Barr virus (EBV; human herpesvirus 4 [HHV-4])**

- discovered and named by M.A. Epstein and Y. Barr
- infectious mononucleosis: fever (lasting 2 weeks), acute pharyngitis (sore throat), lymphadenopathy (swollen glands),
- associated with particular forms of cancer
 - Hodgkin's lymphoma, Burkitt's lymphoma, and nasopharyngeal carcinoma
 - conditions associated with HIV: hairy leukoplakia, CNS lymphomas
- autoimmune diseases: dermatomyositis, rheumatoid arthritis, etc.

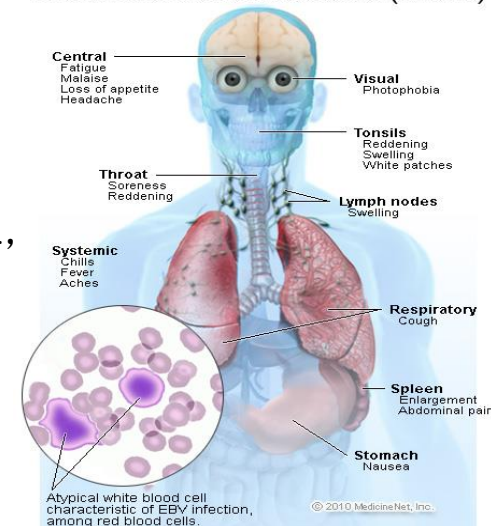


Hodgkin's lymphoma

Burkitt's lymphoma

NC

Infectious Mononucleosis (Mono)



► Family *Herpesviridae*

- Subfamily *Gammaherpesvirinae*

2. Rhadinovirus

- Kaposi's sarcoma-associated herpesvirus (KSHV; human herpesvirus 8 [HHV-8])
- discovered and named by M. Kaposi (in 1872)
- causing a blood vessel tumor (skin cancer)
 - no symptoms in healthy people (latent cycle)
 - immunosuppressed (e.g., AIDS): cancer (lytic cycle)



KS on the palm of the hand. The tumor frequently starts as a raised, discolored, painless lesion that resembles a bruise. This image is from <http://www.pathguy.com/lectures/imm-iv.htm>.

Papovaviridae

Papovaviridae (General properties...)

- Etymology:
Pa- = papilloma, -Po- = polyoma, -Va = vacuolating agent
- Icosahedral in shape
 - Size of virion: 55 nm (Papilloma-) & 45 nm (Polyoma-)
- Non-enveloped, circular dsDNA virus (Group 1)
 - having covalently closed ends
 - supercoiled
- Being resistant to diverse environmental insults:
lipid solvents and detergents, low pH, high temperatures, etc.
- Tumorigenicity; associated with various neoplasm
- Two disparate genera: Papillomavirus and Polyomavirus
(including SV40)
- no longer used in recent taxonomy → split into the *Papillomaviridae* and the *Polyomaviridae*

Papillomaviridae (General properties...)

- History:
 - First identified in the early 20th century
: skin wart-causing filterable infectious agent
 - In 1935, F. P. Rous: papillomavirus causing skin cancer in rabbits
- Etymology : papilla (유두) + oma (종양) → papilloma (유두종)
- Classification: containing 30 genera
- Highly host- and tissue-tropic
 - hosts: mammals (humans, cows, horses, dogs, rabbits, etc.), birds, reptiles, etc.
 - tissues: a particular body surface → skin, mucosal epithelium
- Pathogenicity
 - : generally, being asymptomatic or causing small benign tumors (known as papilloma or wart)

Papillomavirus (Replication)...

- Replication
 - : exclusively in keratinocytes (forming the outermost layers of skin and mucosal surfaces)
- Life cycle
 - Entry: L1 protein vs. alpha-6 beta-4 integrin
 - receptor-mediated endocytosis
 - DNA release & transport: mediated by L2 capsid protein
 - Transcription and DNA replication: in nucleus
 - Early proteins (E1 to E8): regulatory proteins; related to persistence, DNA replication, and activation of the lytic cycle
 - E1 and E2: regulating and maintaining viral DNA as a circular episome
 - E6 and E7: oncogenes – inactivating tumor suppressor proteins (p53)
 - Late proteins (L1 to L2): structural proteins

Papillomavirus (Classification)...

- Human papillomavirus: >170 types
- Bovine papillomavirus: 6 types
- Equine papillomavirus
- Pig genital papilloma virus
- Sheep papillomavirus
- Goat papillomavirus
- Monkey papillomavirus
- Canine papillomavirus
- Canine oral papillomavirus
- Cottontail rabbit papillomavirus
- Rabbit oral papillomavirus

Papillomaviridae (General properties...)

- Human papillomavirus:

- over 170 types have been completely sequenced.
- divided into 5 genera
 - : alpha-, beta-, gamma-, mu-, and nu-papillomavirus
- at least 200 additional viruses have been identified that await sequencing and classification.
- representative types:
 - cutaneous papillomatosis (피부증후군)
 - HPV 1: plantar wart (발바닥사마귀) / warts on feet
 - HPV 2, 4: common wart (보통사마귀) / warts on palms of hands
 - HPV 3, 10: flat wart (편평사마귀) / smooth, flat wart occurring most commonly on the face and legs
 - HPV 5, 8, 17, 20, 36: epidermodysplasia verruciformis (사마귀표피형성 이상) → squamous cell carcinoma (skin cancer)
 - mucocutaneous papillomatosis (점막증후군)
 - HPV 6, 11: oral or laryngeal papilloma, condyloma
 - HPV 16, 18, 45: genital carcinoma (cervical cancer; 자궁경부암)

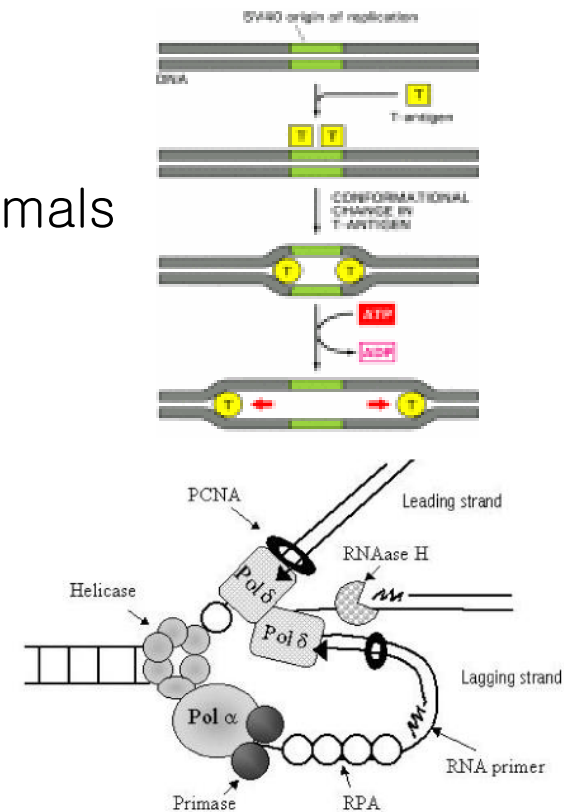
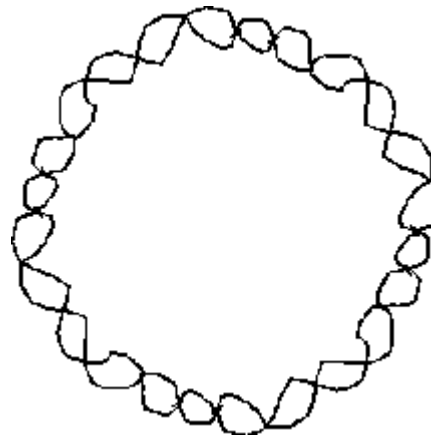
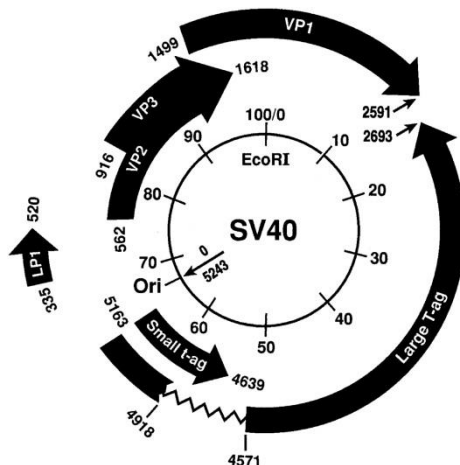


Papillomaviridae (Others...)

- **Transmission:** close contact (via wounds), sexual practices, congenital infection, etc.
- **Diagnosis:**
 - Pap smear (Papanicolaou test): examining the cells from endo-cervical canal under a microscope to look for abnormalities
 - Cervicography (자궁경부사진촬영)
 - PCR, DNA chip using oligonucleotide microarray, etc.
- **Control:**
 - prevention: vaccines (e.g. virus-like particle (VLP) – major capsid protein, L1)
 - Gervarix (GSK): a VLP vaccine against type 16+18
 - Gardasil (MSD): a VLP vaccine against type 16+18+6+11
 - treatment:
 - use of lasers in surgery, cryotherapy, electrotherapy
 - podophyllin solution: the rhizome of American Mayapple
 - imiquimod (Aldara), interferon: immune modulators

Polyomavirus (General properties)...

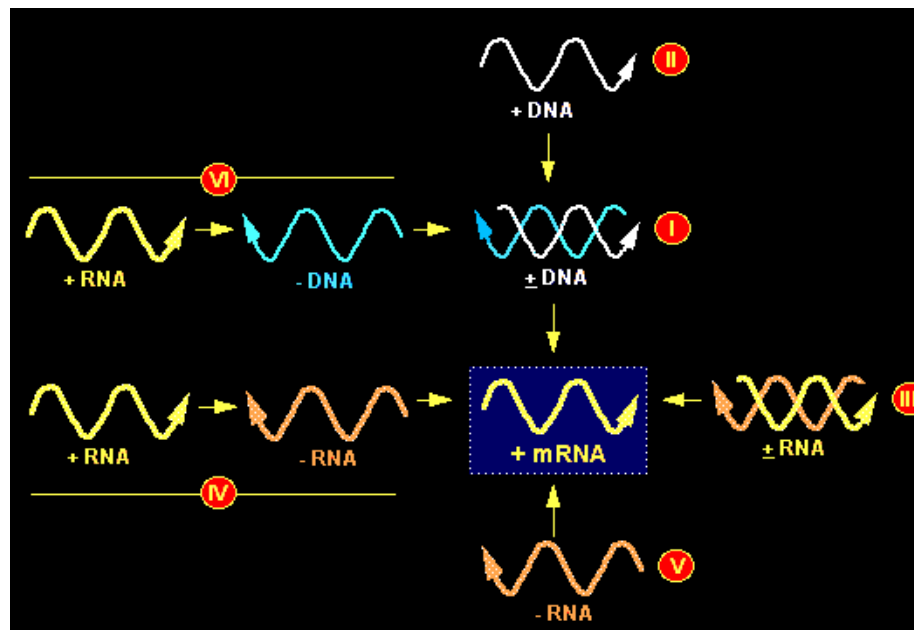
- History:
 - In 1953, Ludwik Gross: first polyomavirus was discovered : Murine polyomavirus
 - In 1975, Renato Dulbecco: insertion of polyomavirus DNA into cellular DNA → “transformation”
- Etymology : poly- = multiple, -oma- = tumor
- Representative tumor viruses in humans and animals
- Simian virus 40 (SV40)



Polyomavirus (Classification)...

- Three major clades
 - : based on genetic characteristics
 - 1. SV40 clade
 - SV40
 - JC polyomavirus (human)
 - BK polyomavirus (human)
 - Simian agent 12 (SV12)
 - 2. Avian clade
 - Avian polyomavirus (Budgerigar Fledgling disease virus)
 - Goose hemorrhagic polyomavirus
 - 3. Murine polyomavirus clade
 - Murine pneumotropic virus
 - Squirrel monkey polyomavirus
 - California sea lion polyomavirus 1

Class 2: ssDNA virus



- Bacteriophage families
 - Family Inoviridae
 - Family Microviridae
- Family **Parvoviridae**
- Family Circoviridae
- (Ad) Families Anelloviridae, Geminiviridae, Nanoviridae

Parvoviridae

Parvoviridae...

- Classification: 2 subfamilies

1. Parvovirinae:

- Parvovirus: murine virus (not infect human)
- **Erythrovirus**: representative parvovirus in human
- Bocavirus: respiratory diseases in bovine, ,canine, porcine, and humans
(causing diseases with very rare rate in humans)
- Dependovirus: Adeno-associated virus; needs a helper virus
(adenovirus, herpesvirus or vaccinia virus);
Infect all vertebrates, but no disease known in human
- Amdovirus: Aleutian mink disease

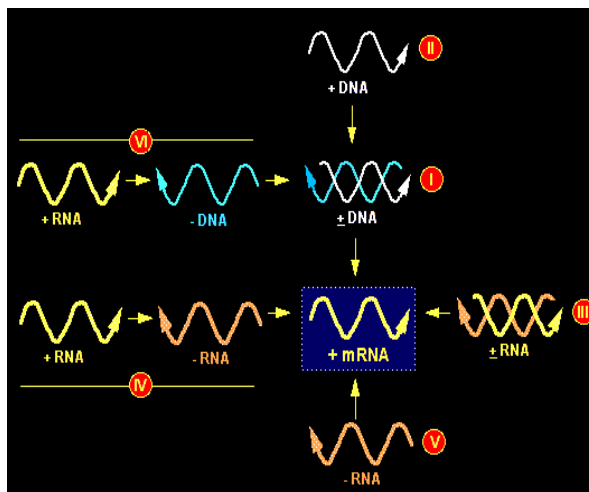
2. Densovirinae: currently not focused

Parvoviridae...

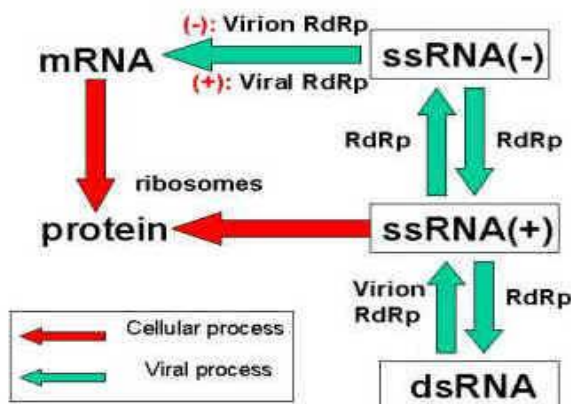
- Erythrovirus:
 - Previously called Parvovirus B19
 - B19 = “discovered in well B19 of petri dishes”
 - Erythro- = “its capability to invade RBC precursors in the bone marrow”
 - Spread: respiratory droplets (+blood-borne transmission)
strong transmissibility (the secondary attack risk of 50%)
 - Symptoms:
 - Erythema infectiosum (“fifth disease”) in children
 - Seronegative arthritis in adults
 - Anemia and related diseases
 - History of “fifth disease”
 - : 1. Measles (홍역), 2. Scarlet fever (성홍열), 3. Rubella (풍진),
4. Duke’s disease, 5. Erythema infectiosum (전염성홍반),
6. Roseola (홍진, 장미진)



Class 3: dsRNA virus



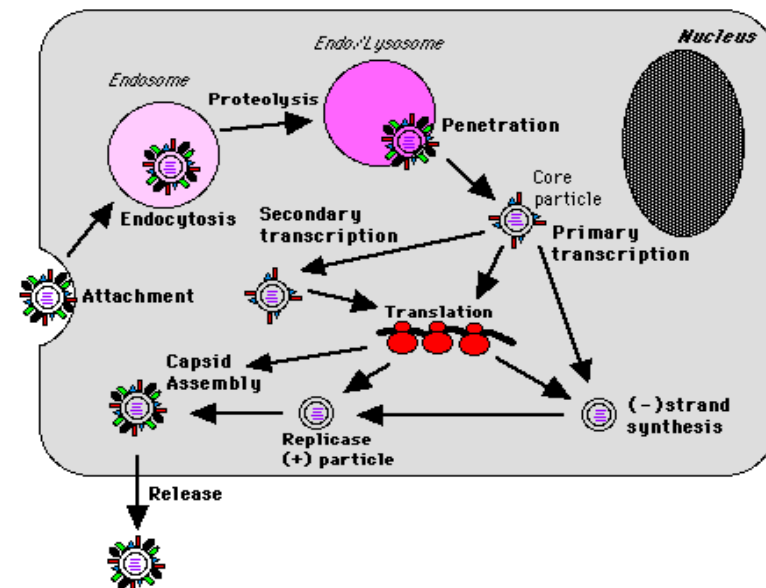
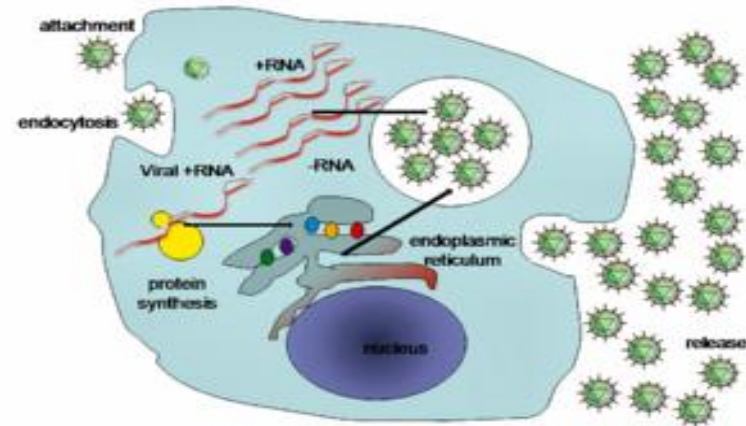
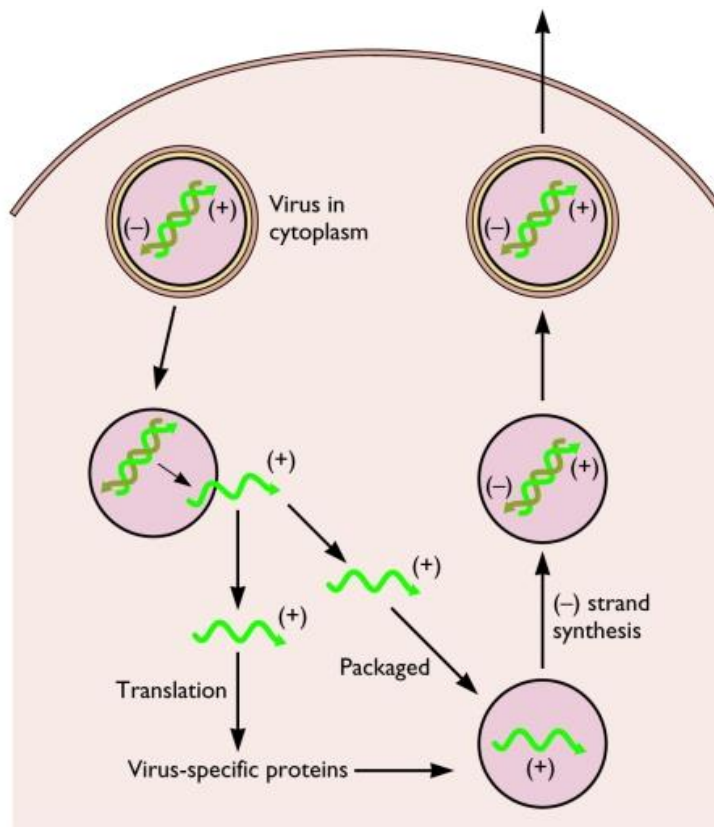
Pathways of Information Flow for RNA Viruses



Conservative replication mechanism :

- Step 1: primary tp / tl
dsRNA → (+) ssRNA → proteins
- Step 2: immature virions
(+) ssRNA + struc proteins
- Step 3: immature virions
dsRNA + struc proteins
- Step 4: secondary tp / tl
dsRNA → (+) mRNA → proteins
- Step 5: final assembly (mature virions)

Class 3: dsRNA virus



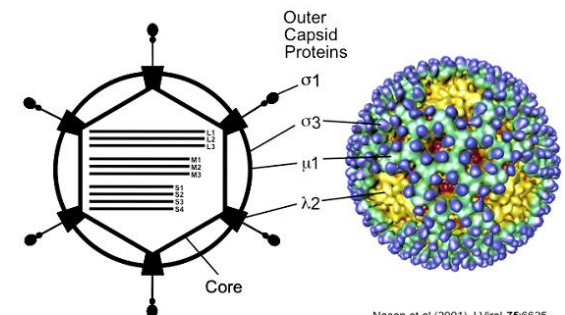
Group 3: dsRNA virus

- Family [Reoviridae](#)
- Family Birnaviridae (fish, bird, insect)
- Family Partitiviridae (plant, fungi)
- Family Hypoviridae (fungi)
- Family Chrysoviridae (fungi – *Penicillium*)
- Family Totiviridae (yeast – *Saccharomyces cerevisiae*)
- Family Cystoviridae (bacteria – *Pseudomonas species*)

Reoviridae

Reoviridae (General properties)...

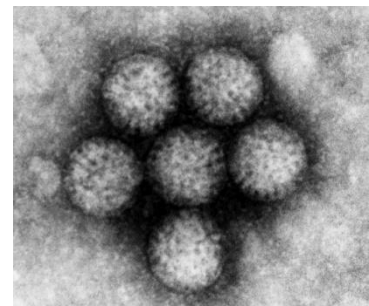
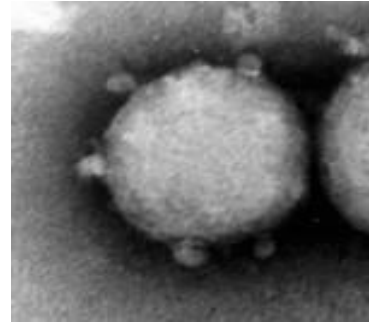
- Reo- = “**R**espiratory **E**nteric **O**rphan (virus)”
(orphan virus = virus not associated with any known diseases)
- Non-enveloped, ds RNA virus (Group 3)
- Segmented genome: 10–12 segments
→ L (large), M (medium), and S (small) groups of segments
- Replication: viral RDRP → cytoplasm
- Affects the gastrointestinal system and respiratory tract
→ Detected in feces, pharyngeal or nasal secretions, urine, cerebrospinal fluid, and blood
- Mild or subclinical infections in humans
- The role of reovirus in clinical infections and treatment are uncertain.



Nason et al (2001) J Virol 75:6625
Figure 1. The reovirus virion. Left panel: schematic diagram of the reovirus virion, denoting the attachment protein, $\sigma 1$ and its $\lambda 2$ base; the outer capsid, composed of proteins $\sigma 3$ and $\mu 1$; and the core. The ten genomic dsRNA segments are shown, and are categorized by relative length: small, or S, segments encode the σ proteins, M segments encode the μ proteins, and L segments encode the λ proteins. Right panel: cryoelectron micrograph reconstruction of the reovirus virion. The trimeric $\sigma 1$ protein is not resolved in the micrograph due to a symmetry mismatch with the pentameric $\lambda 2$ protein.

Reoviridae (Classification)...

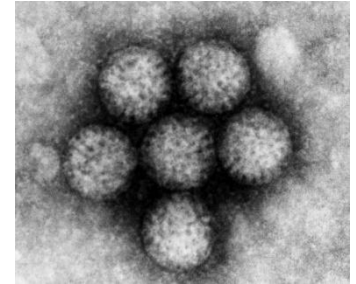
- Two subfamilies
 1. Spinareovirinae
 - : spina- = “spike or turret”
 - Aquareovirus, Coltivirus, Cypovirus, Dinovernavirus, Fijivirus, Idnoreovirus, Mycoreovirus, Orthoreovirus, Oryzavirus
 2. Sedoreovirinae
 - : sedo- = “smooth (the absence of spikes)”
 - **Rotavirus**, Cardoreovirus, Mimoreovirus, Orbivirus, Phytoreovirus, Seadornavirus



Rotavirus (General properties)...

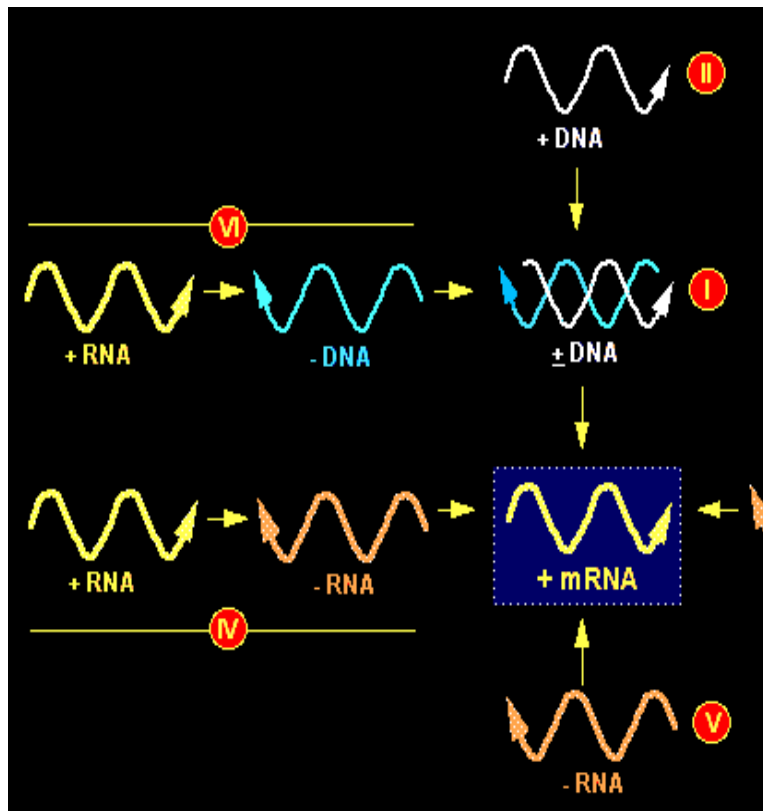
- Zoonosis: humans, other mammals
(apes, cattle, pigs, horses, sheep, dogs, cats, mice, etc.)
- In humans: the most common cause of severe diarrhea among infants and young children
→ gastroenteritis (stomach flu)
- Five species: Rotavirus A (most common), B (adult rotavirus),
C (rare and sporadic cases in children) / D, F, G (birds) /
E, H (pigs) / I (cats)
 - Rotavirus A: the most common in humans (>90% of infections)
 - Children: nearly every child in the world has been infected at least once /
>500,000 children under five years die
 - Adults: be rarely affected, but the elderly and immunocompromised
population can be affected
- Replication: in the cells lining the small intestine

Rotavirus (Others)...

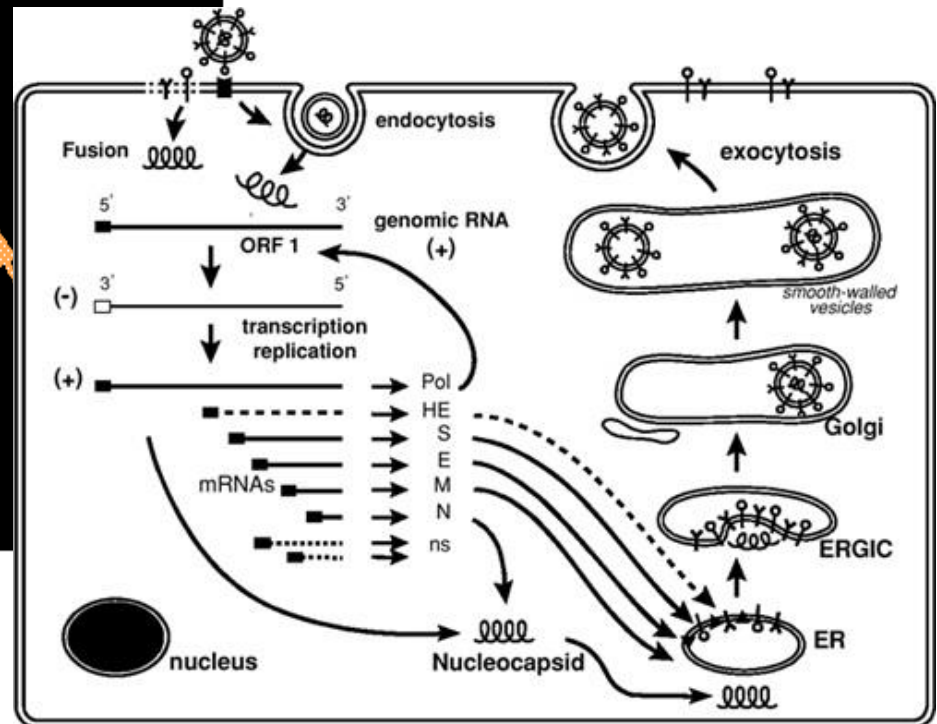


- History
 - 1943: • Jacob Light and Horace Hodes
 - Infectious, filterable agent in the faeces of children with diarrhea
 - causing scours (livestock diarrhea) in cattle
 - 1974: • Thomas Henry
 - viewed it by electron microscopy
 - named it rotavirus (rota- = L. wheel)
- Transmission: the fecal-oral route via contact with contaminants
(possibly by the respiratory spread through aerosol, although not proved in humans)
- Being stable in the environment
- Vaccination: 1998 (first licensed vaccine) → 1999 (withdrew by the manufacturer; intussusception) → 2006 (new vaccines developed) → 2009 (WHO recommended)

Group 4: (+) ssRNA virus



* Most (+) sense RNA genomes are infectious for themselves.



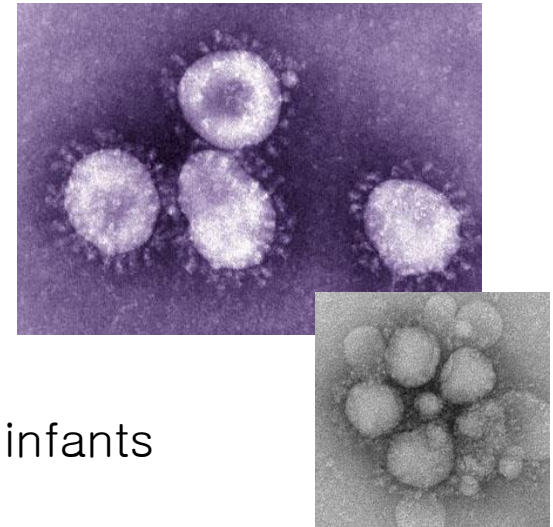
Group 4: (+) ssRNA virus

- Order Nidovirales
 - Family Arteriviridae
 - Family **Coronaviridae**
 - Family Roniviridae
- Order picornavirales
 - Family **picornaviridae**
 - Family Dicistroviridae
 - Family Iflaviridae
 - Family Marnaviridae
 - Family Secoviridae
- Order Tymovirales
 - Family Alphaflexiviridae
 - Family Betaflexiviridae
 - Family Gammaflexiviridae
 - Family Tymoviridae
- Unassigned families
 - Family **Caliciviridae**
 - Family **Flaviviridae**
 - Family **Togaviridae**
 - Family etc.
- Unassigned genera
 - Genus **Hepevirus** (Hepatitis E virus)
 - Genus etc.

Coronaviridae

Coronaviridae...

- Order Nidovirales; Enveloped, (+) ssRNA virus
- Etymology: Corona = G. Crown
- Hosts: mammals and birds
 - Primarily replicate in the respiratory tract
 - Also replicate in the enteric route of very young infants (rarely cause neurological syndromes)
- Symptoms: mostly mild (common cold), but severe (SARS) in humans
- Classification (subfamilies):
 1. Coronavirinae:
 - Alphacoronavirus:
: human (HCoV-229E), feline, porcine, canine virus, etc.
 - Betacoronavirus: human (HCoV-HKU1, SARS-CoV, MERS-CoV), murine virus, etc.
 - Gammacoronavirus: avian virus
 2. Torovirinae: Bafinivirus / Torovirus

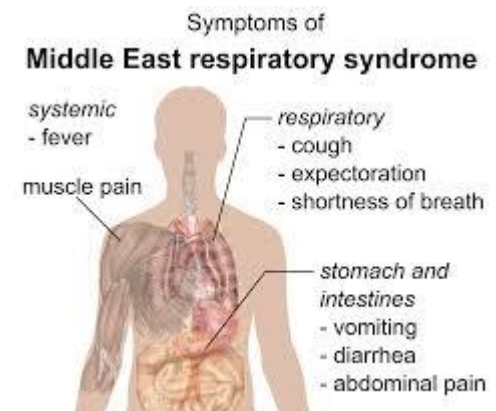


Coronaviridae...

- Diagnosis: in general, diagnosis is not performed
 - RNA detection by RT-PCR
 - serologic assays
- Control:
 - prevention: no commercial vaccines against human coronavirus (animal vaccines only)
 - treatment:
 - early time of SARS outbreak: ribavirin, high concentration of steroids, machine respiration, etc.
 - late time of SARS outbreak: interferon- α , SARS-specific immunoglobulin, lopinavir, ritonavir, etc.
 - be still developed in laboratories
 - quarantine

Middle East respiratory syndrome (MERS)

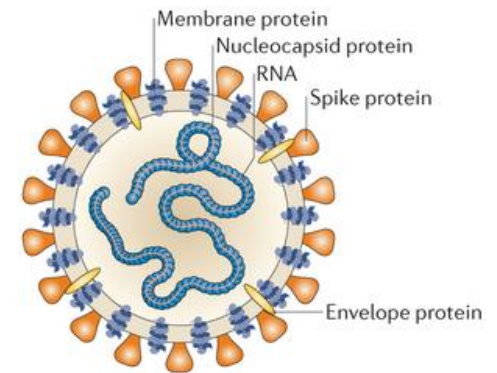
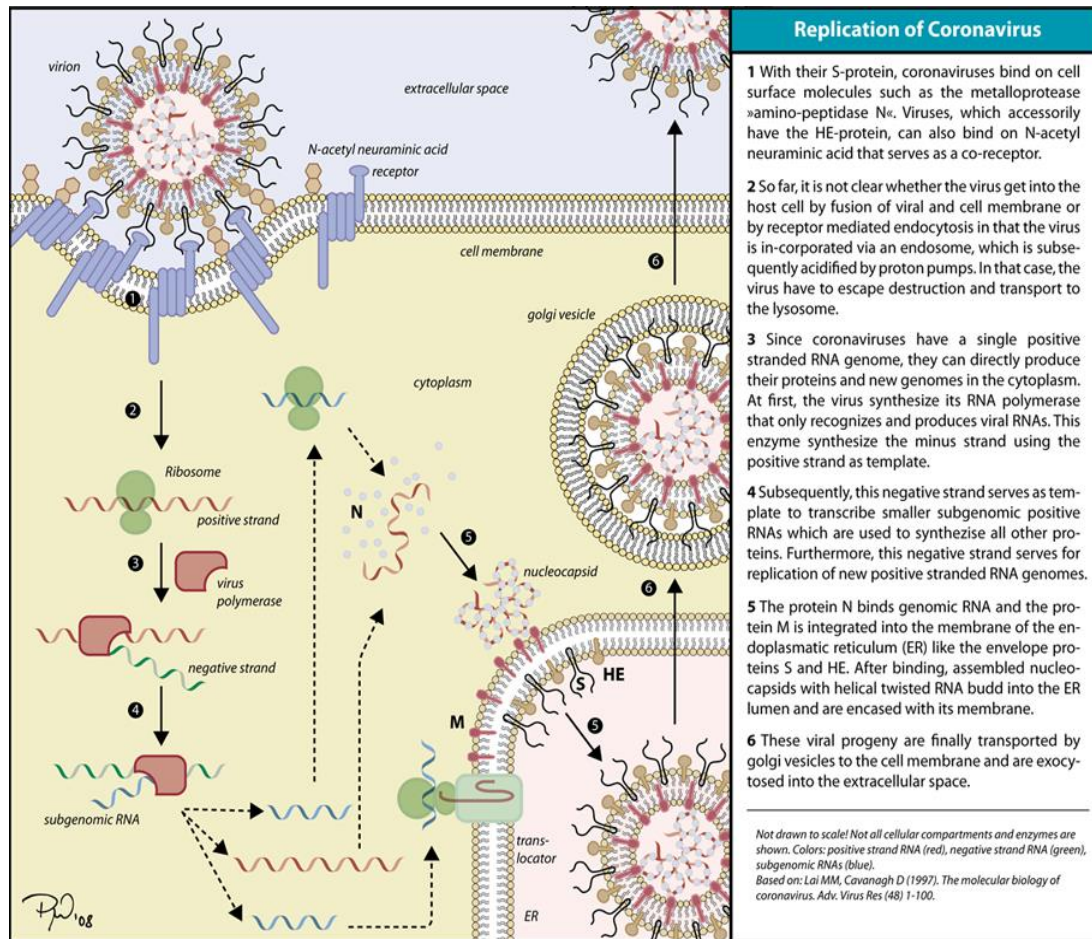
- Viral respiratory disease caused by a novel coronavirus (MERS-CoV)
- First identified in Saudi Arabia in 2012
- Symptoms:
 - ranging from no symptoms (asymptomatic) or mild respiratory symptoms to severe acute respiratory disease and death
 - typical symptoms: **fever, cough and shortness of breath**
 - **pneumonia**: be common, but not always present
 - **gastrointestinal symptoms**, including diarrhea
 - severe illness:
 - causing respiratory failure: requiring mechanical ventilation and support in an intensive care unit
 - more severe in **older people, people with weakened immune systems, and those with chronic disease** such as cancer, lung disease and diabetes
 - lethality: ~ **36%** of reported patients (?)



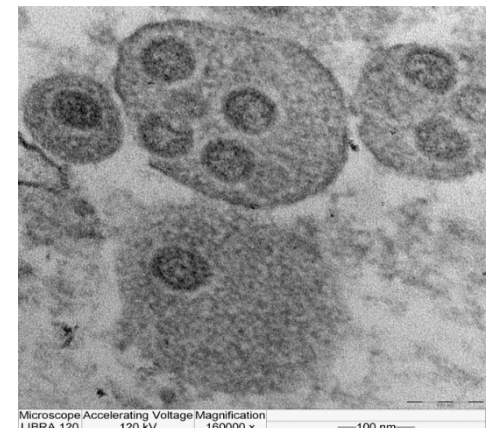
Middle East respiratory syndrome (MERS)

- Being zoonotic:
 - origins: not fully understood, but **probably from bats**
 - intermediate hosts: **camels** – transmitted to camels sometime in the distant past
- Transmission:
 - non-human (animal) to human transmission: not fully understood
 - camels: be likely to be a major reservoir host and animal source of infection in humans
 - human-to-human transmission: not appear to pass easily from person to person unless there is close contact (?)
 - no evidence of sustained human to human transmission
- Prevention and treatment:
 - **no vaccine or specific treatment**
 - treatment: be supportive and based on the patient's clinical condition

Middle East respiratory syndrome (MERS)



Nature Reviews | Microbiology



Piconaviridae

Picornaviridae (General properties)...

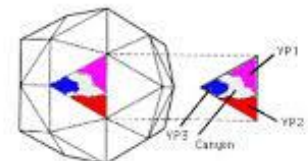
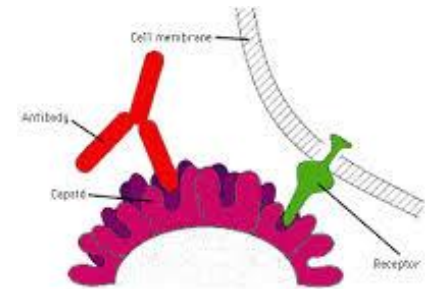
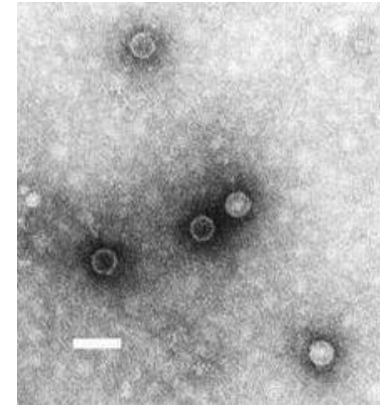
- Non-enveloped, icosahedral
 - stable to heat, detergents, acids, etc.
 - different resistibility depending on species
 - : ex) rhinovirus is only stable to acids.
- (+) ssRNA genome
- Pico- = “small”; -rna- = RNA → “small RNA virus”
- Hosts: humans and animals (livestocks and birds)
- Replication in cytoplasm / infectious naked genome
- Symptoms: being varied (from common cold to poliomyelitis)

Picornaviridae (Classification)...

- 35 genera / 80 species
→ representative: genus *Enterovirus*, *Hepatovirus*, *Aphthovirus*, etc.
- genus *Enterovirus*: species *Enterovirus A* ~ *J* and *Rhinovirus A* ~ *C*
 - *Enterovirus A* : some of Coxsackievirus A, serotype Enterovirus A (EV-As), etc.
 - *Enterovirus B* : Coxsackievirus B, Echovirus, serotype Enterovirus B (EV-Bs), etc.
 - *Enterovirus C* : Poliovirus, some of Coxsackievirus A
 - *Rhinovirus A* ~ *C*
- genus *Hepatovirus* : *Hepatitis A virus*
- genus *Aphthovirus* : *Foot-and-mouth disease virus*, etc.

Picornaviridae (Classification)...

- Several genera: enterovirus, hepatovirus, aphthovirus, etc.
- 1. **Enterovirus**: Enterovirus A~J (including Coxsackie, Echo and Poliovirus), Rhinoviruses (A, B and C)
→ not causing enteric diseases !
- 1) **Poliovirus**: member of enterovirus C (PV-1, PV-2 and PV-3)
 - History:
 - first isolated in 1909: Karl Landsteiner and Erwin Popper
 - full sequencing in 1981: David Baltimore / Naomi Kitamura
 - Host specificity: limited (humans and chimpanzee only)
 - Infection: oral or respiratory tracts
 - Replication:
 - entry: poliovirus “canyon” vs. CD155
→ injection of viral RNA only or receptor-mediated endocytosis
 - all processes occur in cytoplasm



Picornaviridae (Classification)...

- Several genera: enterovirus, hepatovirus, aphthovirus, etc.

1. Enterovirus:

1) Poliovirus: member of Enterovirus C (PV-1, PV-2 and PV-3)

– Symptoms:

- asymptomatic (95%): caused by primary, transient viremia
- mild symptoms (5%): caused by replication and spreading to other sites of tissues, followed by secondary viremia
→ fever, headache, sore throat, etc.
- severe symptoms (<1%): caused by CNS infection
→ temporary or permanent paralysis (paralytic poliomyelitis)

– Epidemiology:

- transmitted by close contacts (via respiratory droplets), feces (oral infections via food or water intake or in swimming pools), etc.
- prevalence in late summer to winter

– Diagnosis: virus isolation in cell culture, PCR, etc.

– Control: vaccines (killed/live-attenuated) / no specific treatments

Picornaviridae (Classification)...

- Several genera: enterovirus, hepatovirus, aphthovirus, etc.

1. Enterovirus:

2) **Coxsackie viruses:** members of either Enterovirus A, B or C

- Etymology: first isolation at Coxsackie in NY
- Classification:

- Group A:

- 24 serotypes (A1 to A24)
- hand, foot and mouth disease (a common childhood illness)
 - often produced by coxsackie A16 (+A4, 5, 9 and 10)
 - in most cases, asymptomatic or mild symptoms
 - affecting mostly children aged 5 or under (+can also affect adults)
 - fever, sore throat, itchy and painful rash and blisters on hands and feet and in mouth
 - transmission: nasal discharge, saliva, faeces & fluids from blisters
- other diseases: acute hemorrhagic conjunctivitis (A24 specifically), herpangina (포진성구협염, 구내염), aseptic meningitis, etc.



Picornaviridae (Classification)...

2) Cocksackie viruses: members of either Enterovirus A, B or C

– Classification:

· Group B:

– 6 serotypes

– diseases:

· (epidemic) pleurodynia (유행성홍막통(염); Bornholm disease)

– fever, headache, sore throat, gastrointestinal distress, chest and muscle pain) etc.

– pericarditis or myocarditis → permanent heart damage or death (~50% of sudden death cases)

· HFMD (coxsackie B2 and B5), etc.

– Transmission: fecal–oral route (after eating contaminated food) or respiratory tract

– Diagnosis: mainly via serological tests (ELISA) and from PCR, etc.

– Control: no specific treatments and no vaccines

Picornaviridae (Classification)...

- Several genera: enterovirus, hepatovirus, aphthovirus, etc.

1. Enterovirus:

3) Echovirus: members of *enterovirus B*

- Echo– = acronym for “enteric cytopathic human orphan”
- 32 serotypes
- Highly infectious; primarily targets children (or adult males)
- Symptoms:
 - typically, asymptomatic or mild illness (e.g. fever, rash, etc.)
 - potentially fatal disease in infants
 - myocarditis in the most frequent complication in adults
 - aseptic meningitis: the most common cause of it
- Transmission: alimentary (e.g. contaminated food and water; swimming pool) and respiratory (e.g. saliva) tracts / hospital-acquired
- Diagnosis: virus isolation in cell culture
- Control: no vaccines / pleconaril

Picornaviridae (Classification)...

- Several genera: enterovirus, hepatovirus, aphthovirus, etc.

1. Enterovirus:

4) Human rhinovirus

- Rhino– = G. nose
- 105 serotypes; no cross-reactivity of immunity
- Infection: occurs primarily in the nose
(by aerosols of respiratory droplets and from contaminants)
- Replication: proliferates in temperatures b/w 33–35 degree
- Symptoms: common cold (35–50% of common cold; sore throat, runny nose, nasal congestion, sneezing, cough, muscle aches, head aches, etc.) → secondary infection of bacteria
- Transmission: respiratory droplets
- Control:
 - treatment: IFN- α , pleconaril, Iota-carrageenan
 - vaccination: currently no vaccines available (but, VP4 is highly conserved)

Picornaviridae (Classification)...

- Several genera:

1. Enterovirus: poliovirus, coxsackie virus, echovirus, rhinovirus

2. Hepatovirus: **Hepatitis A virus**

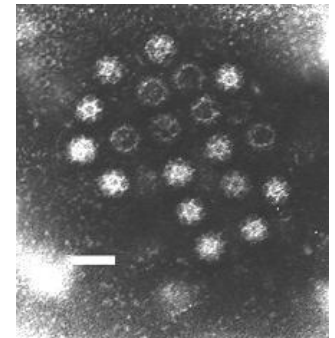
- Infection: occurs via the fecal–oral route
(person–to–person by ingestion of contaminants)
- Be stable to acid and ether, but not to heat and chemicals (chlorine, formalin, etc.)
- Symptoms:
 - asymptomatic infection: >90% of infected children
→ conferring lifelong immunity
 - symptomatic infection: >80% of adults
 - early symptoms of hepatitis A: similar as those of influenza
 - acute form of hepatitis (not having a chronic stage)
 - less than hepatitis B
- Diagnosis: serologic tests (e.g. ELISA)
- Control: vaccination and good hygiene / no specific treatments

3. Aphthovirus: **Foot–and–mouth disease virus (FMDV)**

Caliciviridae

Caliciviridae...

- Calici- = L. calyx (“cup or goblet”)
- Small (27–38 nm); icosahedral
- Non-enveloped: less susceptible to alcohols and detergents, but rapidly inactivated by sufficient heating and chlorine
- Not grow in culture → not very well studied
- Classification: 4 genera – Norovirus (formerly Norwalk agent; humans), Sapovirus, Vesivirus, Lagovirus
- Hosts: humans, cattle, pigs, cats, dolphins, chickens, reptiles and amphibians, etc.
- Transmission: mainly through fecal–oral route from contaminated water and food (especially, shell–fish)



Caliciviridae...

- Symptoms:
 - major cause of gastroenteritis
 - nausea, vomiting, watery diarrhea, abdominal pain, loss of taste, lethargy, muscle ache, headache, cough, low-grade fever, etc.
 - “winter vomiting bug” in the UK
 - last for only 3 days; do not call for medical attention
- Diagnosis: PCR or ELISA (low specificity and sensitivity)
- Control:
 - no vaccines
 - no specific antiviral drugs
 - allopathy: rehydration therapy for immunocompromised patients

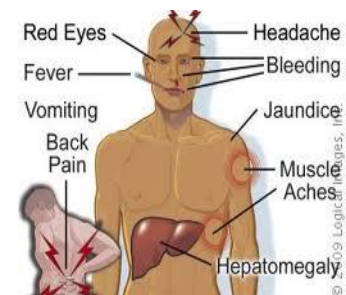
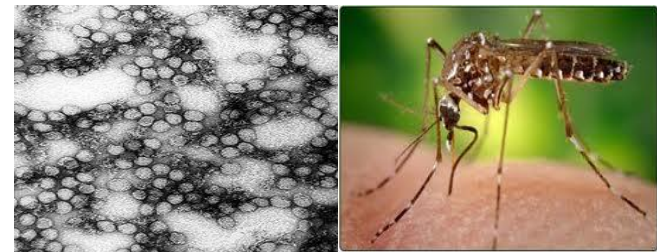
Flaviviridae

Flaviviridae...

- Flavi- = L. flavus (“yellow”; due to cause jaundice in victims / the family name got from yellow fever virus)
- Enveloped; icosahedral (37–75 nm)
- Previously, a member of the family *Togaviridae*
- Classification: 4 genera
 1. Genus *Flavivirus*: **Yellow fever virus, West Nile virus, Dengue fever virus, Japanese encephalitis virus, Zika virus**
 2. *Hepacivirus*: **Hepatitis C virus**
 3. *Pegivirus*: GB virus C (formerly Hepatitis G virus)
 4. *Pestivirus*
- Arbovirus: a term used to describe viruses transmitted by arthropod vectors / “Arbo-” is an acronym of **AR**thropod-**BO**rne.
→ cf. *tibovirus*: **TI**ck-**BO**rne virus
- Diagnosis: cytopathologic examination (세포병리검사), immunological tests, hemagglutination assay, RT-PCR, etc.
- Control: vaccines (in some viruses) / no specific treatments

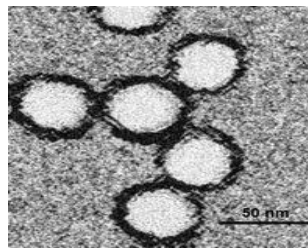
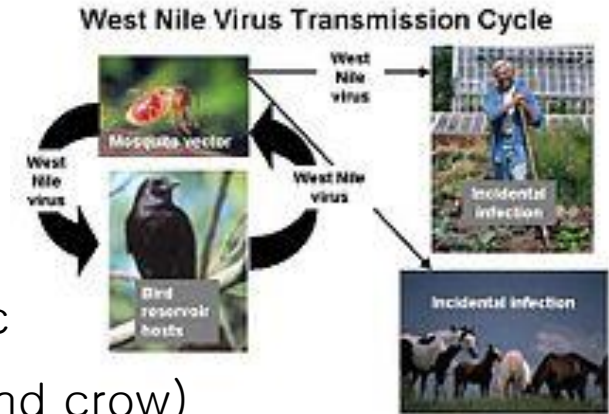
Yellow fever virus

- Hosts: primates / zoonotic
- Vectors: female mosquitoes
(yellow fever mosquito, *Aedes aegypti*, etc.)
- Symptoms: an acute hemorrhagic disease / reemerging since 1980s
 - mild signs: fever, headache, chills, nausea, vomiting, etc.
 - severe signs: liver damage with jaundice, bleeding diathesis, and death
 - WHO: 200,000 infections and 30,000 deaths / year
 - one of the most dangerous infectious diseases
- Distribution:
 - tropical and subtropical areas in Africa
 - S. America: 16th century (slave trade)
 - Several epidemics in the 17th century
: Africa, America and Europe
- Controls: vaccination (live-attenuated; side effects
 - e.g. meningoencephalitis) / no therapy



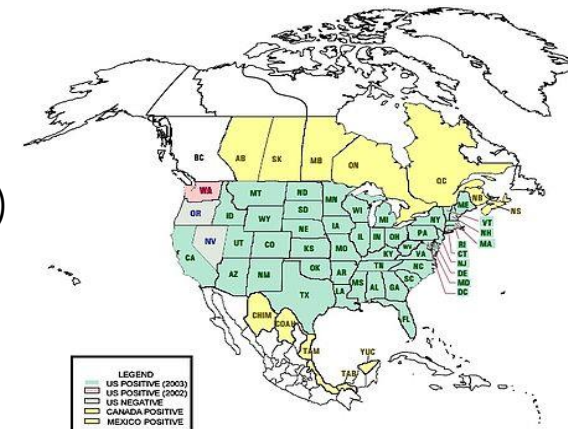
West Nile virus

- Hosts: humans, horses, dogs, cats, bats, chipmunks, skunks, squirrels, rabbits, crocodiles, and alligators / zoonotic
- Natural / amplifying hosts: birds (American robin and crow)
- Vectors: mosquitoes (*Aedes albopictus*, *Culex pipiens*, *Culex tarsalis*, *Culex quinquefasciatus*)
- Symptoms:
 - Asymptomatic infection (80%)
 - Mild febrile syndrome: West Nile fever (20%)
 - fever, headache, chills, diaphoresis, pains: similar as those of influenza
 - Neuroinvasive disease: West Nile meningitis or encephalitis (<1%)
 - coma, death



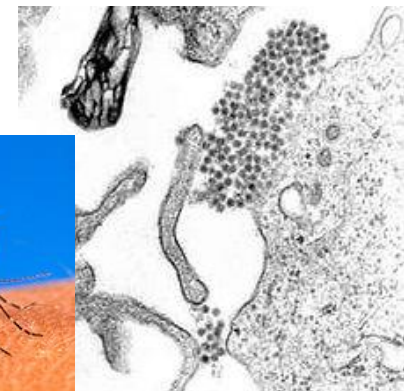
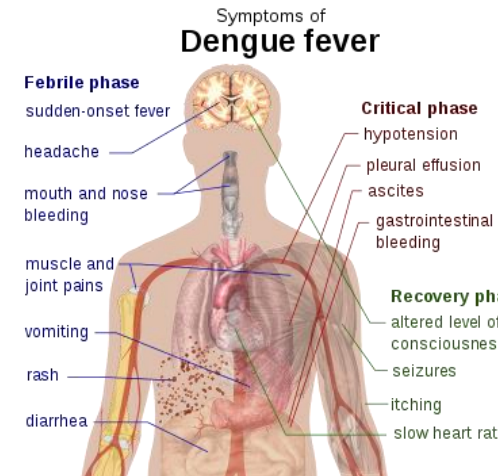
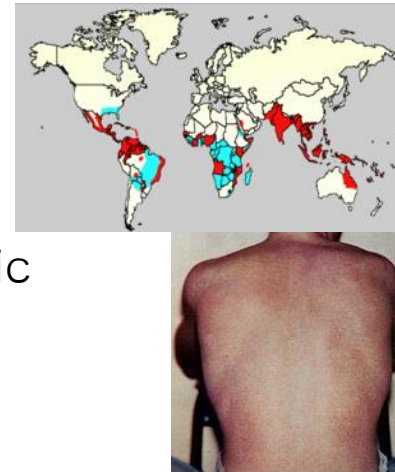
West Nile virus

- Transmission:
 - Human-to-human transmission: caused by occupational exposure or conjunctival exposure to infected blood / blood transfusion, organ transplant, intrauterine exposure, and breast feeding
 - No evidence of pet (dog or cat)-to-human transmission
- Distribution:
 - Africa: emerged around 1,000 years ago (first isolated in the West Nile District of Uganda in 1937)
 - Europe and the US
- Control:
 - no vaccine for humans (but a vaccine for horses)
 - no specific treatments
 - insect repellents



Dengue fever virus

- Four major serotypes: DENV-1 to -4
- Hosts: humans and primates / zoonotic
- Vectors: mosquitoes – *Aedes* genus (principally *Aedes aegypti*)
- Symptoms: an infectious tropical disease
 - commonly asymptomatic or mild-symptomatic infection
 - mild signs: fever, headache, muscle and joint pains, skin rash, etc.
 - severe signs: **dengue hemorrhagic fever** (bleeding, low level of blood platelets, and blood plasma leakage) and **dengue shock syndrome** (dangerously low blood pressure)
 - 50–100 million people infected yearly → 500,000 people with severe dengue
- Distribution:
 - Currently endemic in >110 countries
- Controls: vaccine (live-attenuated tetravalent vaccine; Dengvaxia), but no specific treatment



Zika virus

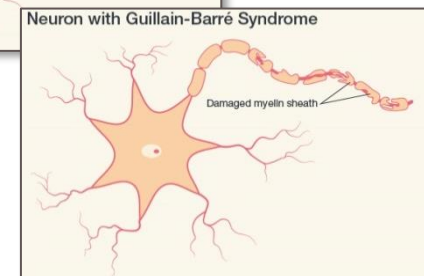
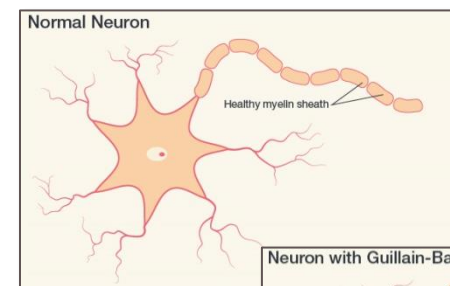
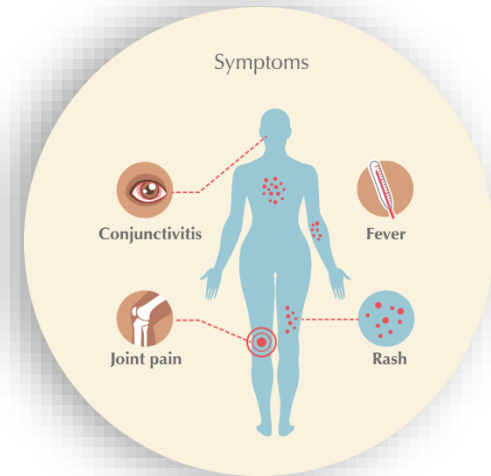
- History :

- first isolated in 1947 from a monkey in the Zika Forest of Uganda
- sporadic outbreaks within a narrow equatorial belt from Africa to Asia (including Malaysia and Indonesia)
 - for ~60 years: since 1947 to 2007
 - 14 cases of human infection, but no inpatients and no deaths
- spreading eastward of the virus
 - 49 confirmed cases of human infection / seropositive to 73% of the population in Micronesia in 2007
 - ~30,000 (10% of the population) confirmed cases of human infection / inpatient cases in Polynesia in 2013–14
- 2015–16 Zika virus epidemic (pandemic levels) in South America
 - beginning in Brazil
 - ~40 countries



Zika virus

- **Symptoms:** Zika fever (Zika virus disease)
 - predominantly **no or only mild symptoms** (a very mild form of dengue fever and chikungunya)
 - fever, conjunctivitis (red eyes), joint pain, skin rash + headache, muscle pain, etc.
 - generally lasting less than seven days
 - **Guillain-Barre syndrome (GBS):** an uncommon sickness of the nervous system / damage of the nerve cells by the immune system → muscle weakness and paralysis
 - **microcephaly:** caused by congenital infection (passing of virus from an infected pregnant woman to her fetus)



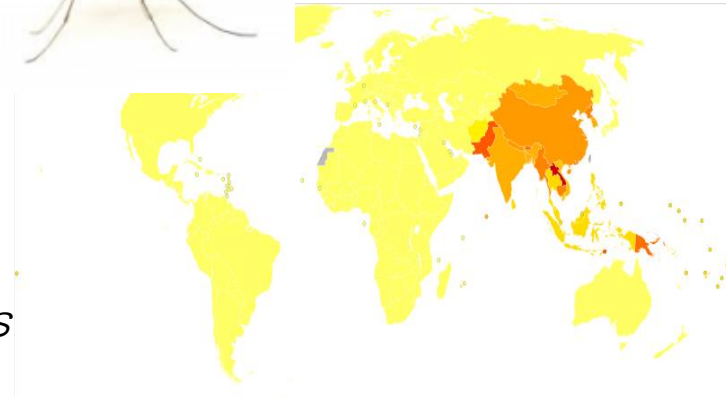
Zika virus

- **Main reservoirs (hosts):** nonhuman (monkeys) and human primates
- **Primary vectors:** mosquitoes – *Aedes* genus (principally *Aedes aegypti*)
- **Modes of transmission:** bite of infectious mosquitoes (mainly *Aedes* spp. mosquitos), blood transfusion, sexual transmission, etc. (but, not by breast feeding–controversial)
- **Incubation period:** estimated to be 3–14 days
- **Diagnosis:** suspected based on symptoms and recent history of travel / confirmed through laboratory tests on blood or other body fluids (e.g., urine, saliva or semen)
- **Viability:** susceptible to common disinfectants (e.g., 70% ethanol, 1% sodium hypochlorite, 2% glutaraldehyde, ether, and temperature >60°C)
- **Controls:**
 - no vaccine and no specific treatment available
 - insect repellents / pain relievers (e.g., acetaminophen, but not aspirin), etc.



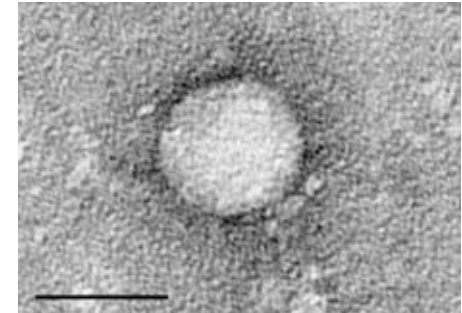
Japanese encephalitis virus

- Hosts: Domestic pigs and wild birds
→ Humans / zoonotic
- Vectors: mosquitoes – *Culex tritaeniorhynchus*
- Symptoms:
 - Commonly asymptomatic infection
 - Only 1 in 250 infections develop into encephalitis.
→ fever, headache, malaise, neck rigidity, cachexia, hemiparesis, convulsions, coma, death
- Distribution:
 - Most prevalent in Southeast Asia and the Far East
- Controls: vaccine (killed) available, but no specific treatment



Hepatitis C virus

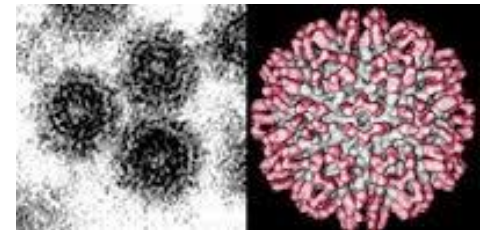
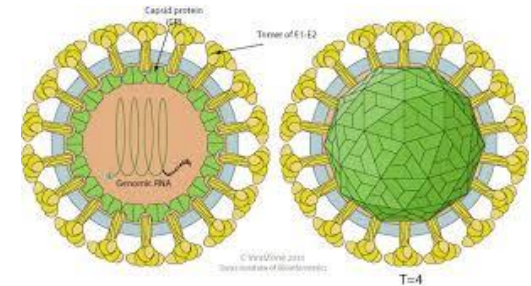
- Hosts: humans and chimpanzees
- Six genotypes with several subtypes
- Replication: mainly in the hepatocytes of the liver /
also in peripheral blood mononuclear cells and B cells
- Symptoms:
 - often asymptomatic
 - chronic infection (50–80% of infected persons) can lead to scarring of the liver (hepatitis C) → liver failure, cirrhosis and liver cancer (hepatitis C is the primary cause of liver cancer)
- Transmission: intravenous drug use, blood transfusions, unsafe medical procedures, sexual practices, etc.
- Diagnosis: immunological assay (ELISA), RT-PCR, etc.
- Controls:
 - no vaccine
 - polyethylene glycol (PEG)–linked (pegylated) IFN- α (peginterferon), ribavirin or a combination therapy of both



Togaviridae

Togaviridae (General properties)...

- Enveloped; icosahedral (65–70 nm)
- Classification: two genera (*alphavirus* and *rubivirus*)
→ included what are now called the Flaviviruses within the genus *Alphavirus*
- Replication:
 - very fast, taking around 4 hours
 - glycoprotein spikes vs. receptors (unknown)
 - occurs in cytoplasm
 - having viral mRNA resembling cellular mRNA
 - : 5'–methylated cap and 3'–poly (A) tail
 - two characteristic rounds of translation
 - 1st : encoding of a polyprotein → self cleavage → nonstructural proteins (responsible for gene expression and replication)
 - 2nd : formation of a sub-genomic fragment, encoding the structural proteins and a (–) sense fragment
 - Assembly and budding



Togaviridae (genus Alphavirus)...

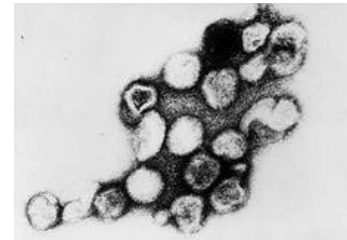
- Hosts: various vertebrates (humans, rodents, fish, birds, horses, etc.) and invertebrates / zoonotic
- Medically important alphaviruses: Western equine encephalitis (서양말뇌염) virus, Eastern equine encephalitis (동양말뇌염) virus, Venezuelan equine encephalitis (베네주엘라말뇌염) virus, Sindbis virus, Barmah Forest virus, Chikungunya virus, etc.
- Transmission: occurring mainly via mosquitoes
- Symptoms in humans:
 - depending on species
 - encephalitis, rash, arthritis, etc.
- Diagnosis: virus isolation in cell culture
- Control: no vaccines and no specific treatments
→ vector control with repellents, protective clothing, etc.



Culiseta

Togaviridae (genus Rubivirus)...

- One species: **rubella virus** (풍진바이러스) → rubella = 'little red' (L.)
- Two genotypes (based on the sequences of the E1 protein): 1 and 2
→ sub-genotypes: 1A to 1J / 2A to 2C
- Symptoms: rubella (also known as German measles or three-day measles)
 - acquired rubella: a common children infection
 - be often mild and attacks often pass unnoticed / can last one to three days (swollen glands or lymph nodes, rash, etc.)
 - children recover more quickly than adults (rarely be fatal)
 - adults: transient arthropathy may occur
 - congenital rubella syndrome (CRS):
 - transplacental infection on the developing fetus
 - occurs during the first weeks of pregnancy
 - cardiac, cerebral, ophthalmic and auditory defects; prematurity; abortion, etc.
- Transmission: close contacts (airborne droplet, urine, feces, etc.)
- Diagnosis: serological tests (Ab detection)
- Control: vaccine (live-attenuated) / no specific treatment



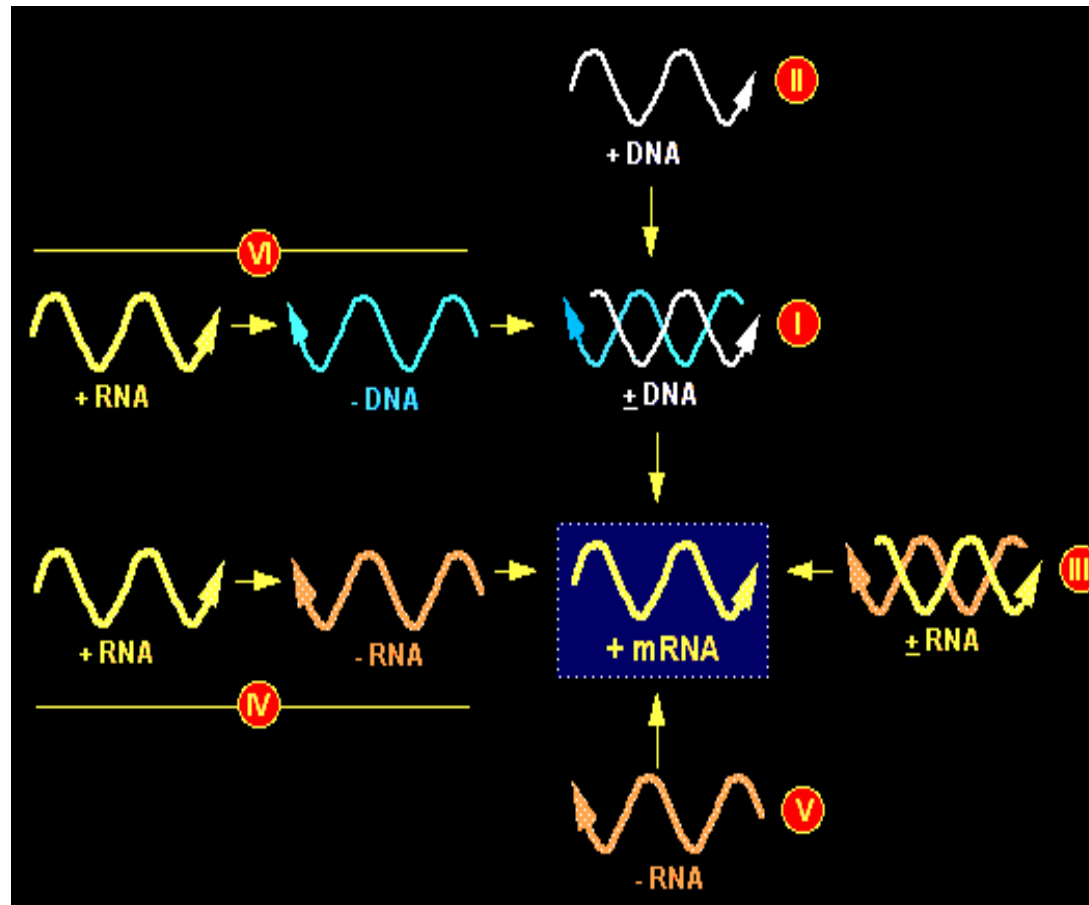
Hepeviridae

Hepeviridae...

- Genus *Hepevirus* / species *Hepatitis E virus*
- Previously classified in the family *Caliciviridae*
- 4 genotypes (1 to 4) in humans; further classified into subtypes → only one serotype
- Hosts: mainly domestic animals (especially domestic pigs), rats, chickens and humans
- Infection: the fecal–oral route; via contaminated foods and water (“E” = enteric)
- Symptoms: usually acute, but may cause chronic (fibrosis, cirrhosis)
 - Commonly asymptomatic infection (+ jaundice, fatigue, and nausea)
 - Occasionally develops into an acute severe liver disease (fatal in about 2% of all cases)
 - In pregnant women, more often severe (elevated mortality rate of 20%)
- Distribution:
 - Prevalent in most developing countries
- Controls: vaccine (recombinant subunit) / no specific treatment



Group 5: (–) ssRNA virus



Group 5: (-) ssRNA virus

- Order Mononegavirales
 - Family Bornaviridae
 - Family Filoviridae
 - Family **Paramyxoviridae**
 - Family **Rhabdoviridae**
- Unassigned families
 - Family Arenaviridae
 - Family **Bunyaviridae**
 - Family **Orthomyxoviridae**
- Unassigned genera
 - Genus Deltavirus
 - Genus Nyavirus
 - Genus Tenuivirus
 - Genus Varicosavirus

Paramyxoviridae

Paramyxoviridae...

- Etymology: (G.) Para- = “beyond”; (G.) -myxo- = “mucus”
- Enveloped / spherical, filamentous or pleomorphic
- Genome: non-segmented (-) ssRNA
- Hosts: humans and animals
- Classification:
 - Subfamily *Paramyxovirinae* → subfamily *Avulavirinae*, *Orthoparamyxovirinae*, *Metaparamyxovirinae* and *Rubulavirinae*
 - Genus *Avulavirus* → subfamily *Avulavirinae* : genus *Ortho-* (NDV), *Meta-* and *Paraavulavirus*
 - Genus *Henipavirus* : member of subfamily *Orthoparamyxovirinae*
 - Genus *Morbillivirus* : member of subfamily *Orthoparamyxovirinae*
 - Genus *Respirovirus* : member of subfamily *Orthoparamyxovirinae*
 - Genus *Rubulavirus* → subfamily *Rubulavirinae* : genus *Orthorubulavirus* (Mumps) and *Pararubulavirus*
 - Subfamily *Pneumovirinae* → family *Pneumoviridae* (2 genera and 5 species)
 - Genus *Orthopneumovirus* : human (RSV), murine, and bovine orthopneumovirus
 - Genus *Metapneumovirus* : avian and human metapneumovirus

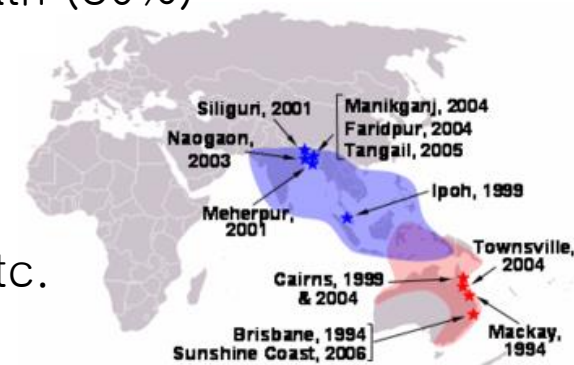
Paramyxovirinae (Avulavirus)

- Classification:
 - **Newcastle disease virus** (currently, Avian ortho-avulavirus 1)
 - Avian paramyxovirus type 2 through 9
 - Goose paramyxovirus
- **Newcastle disease virus (NDV):**
 - A contagious bird disease: domestic and wild birds
: asymptomatic to highly pathogenic (90% mortality)
transmitted through contact with the bodily discharges of infected birds
respiratory signs, nervous signs → reduced egg production and death
 - Endemic in many countries
 - First isolated in Newcastle, UK in 1926
 - Zoonotic: mild conjunctivitis and influenza-like symptoms in humans
 - Previously being one of biological weapons
 - Anticancer agent: selectively kill human tumor cells with limited toxicity to normal cells
 - Vaccines available for animals / no specific treatment



Paramyxovirinae (Henipavirus)

- Representative species: **Hendravirus**, **Nipahvirus**, etc. (currently, Hedra / Nipah henipavirus)
- Natural reservoir: Pteropid fruit bats (flying foxes)
- Zoonotic: causing illness and death in domestic animals and humans
 - In animals:
 - **pigs** (by nipahvirus): porcine respiratory (“mile cough or barking pig syndrome”) and neurologic syndrome → high morbidity but low mortality
 - **horses** (by hedravirus) influenza-like illness and death (80%)
 - In humans:
 - **nipahvirus: respiratory** (e.g., fever, headache, drowsiness, cough), abdominal pain, nausea, vomiting, **neurologic** (e.g., **encephalitis**, seizures), etc. → average mortality rate of 74.5% (up to 100%)
 - **hendravirus: respiratory** (e.g., influenza-like, pneumonic illness, neurologic (e.g., aseptic meningitis, **encephalitis**), etc. → mortality rate of 70%



Paramyxovirinae (Morbillivirus)

- Etymology: Morbilli- = “measles”
- Classification: Canine distemper virus, Cetacean Morbillivirus, **Measles virus**, Peste-des-petits-ruminants virus, Porcine distemper virus, Rinderpest virus
- **Measles virus** (currently, measles morbillivirus):
 - causing an infection of the respiratory system
→ fever, cough, runny nose, Koplik spot, (maculopapular and erythematous) rash, red eyes (blindness), pneumonia, encephalitis (more severe in adults than children) / mortality rate of 3–6% (84% drop in deaths by vaccination)
 - highly contagious / transmitted by contact with fluids from an infected person’s nose and mouth (aerosol transmission)
 - Control: vaccine (MMR – measles, mumps and rubella), but no specific treatment



Paramyxovirinae (Respirovirus)

- Classification:
 - **Human parainfluenza viruses 1 and 3** (currently named human respirovirus 1 and 3)
 - Sendai virus, etc.
- Sendai virus (murine parainfluenza virus type 1):
 - causing a highly transmissible respiratory infection
: sneezing, respiratory distress, lethargy, anorexia
 - transmitted by air and direct contact routes
 - hosts: mice, hamsters, guinea pigs, rats, pigs
- **Human parainfluenza viruses 1 and 3:**
 - causing respiratory illness (cold and sore throat) in younger children (75% of the cases of croup)
 - hPIV-1: most common cause of croup
 - hPIV-3: associated with bronchiolitis and pneumonia
 - no vaccines and no specific treatments available

Paramyxovirinae (Rubulavirus)

- Classification:
 - Mumps virus
 - Human parainfluenza viruses 2 and 4 (currently named human rubulavirus 2 and 4)
 - Simian parainfluenza virus 5
 - Menangle virus, Tioman virus, Tuhokovirus 1,2 and 3
 - etc.
- Mumps virus (currently, mumps orthorubulavirus):
 - causing mumps
 - : fever, headache, swelling of the parotid glands and other epithelial tissues, etc. → complications: meningitis, encephalitis, orchitis, deafness, 25% incidence of abortion, etc.
 - high morbidity, but very low mortality rate (0.001%)
 - transmitted by direct contact, droplet spread or contaminants
 - hosts: natural infection is currently restricted to humans
 - diagnosis: serologic tests (e.g. ELISA, immunofluorescence, HI)
 - control: vaccine (MMR) / no specific treatment



Pneumovirinae (Pneumovirus)

- Classification:
 - Human respiratory syncytial virus (currently human orthopneumovirus)
 - Bovine respiratory syncytial virus
- Human respiratory syncytial virus:
 - etymology: syncytial = “forming syncytia” (by viral F proteins)
 - causing respiratory tract infections
 - mild symptoms: common cold (by upper respiratory infection)
 - severe symptoms: bronchiolitis and pneumonia (under 1 year of age)
 - rarely causing death (by low respiratory infection)
 - 30% of young patients with pneumonia
 - 60% of patients with low respiratory infection
 - transmitted by direct contact, droplet spread or contaminants
 - hosts: humans
 - diagnosis: immunologic tests (e.g. ELISA, immunofluorescence), virus isolation in cell culture, etc.
 - control:
 - no vaccines, but prophylactic drug (Ab; palivizumab) available
 - treatment: ribavirin (a guanosine analog; a nucleoside inhibitor)

Pneumovirinae (Pneumovirus)

- Classification:
 - Human respiratory syncytial virus (currently human orthopneumovirus)
 - Human metapneumovirus, etc.
- Human metapneumovirus:
 - isolated for the first time in 2001 in the Netherlands
 - associated with **6–40% of respiratory tract infections (upper or lower) in hospitalized and outpatient children / the second most common cause after human orthopneumovirus (RSV) of lower respiratory infection in young children**
 - peak age of hospitalization for infants with HMPV: 6–12 months of age (slightly older than the peak of RSV, which is around 2–3 months)
 - also an important **cause of disease in older adults (>65 years)**
 - clinical features and severity of HMPV: similar to those of RSV
 - transmission: no conclusive studies to date / may transmitted by droplet, aerosol, fomite vectors or hospital acquired infections
 - control: no vaccines and no antiviral agents (ribavirin showing effectiveness in an animal model)

Rhabdoviridae

Rhabdoviridae...

- Etymology: Rhabdo- = (G.) rhabdos (“rod”)
- Morphology: rod (bullet-shaped or cylindrical); capsid of helical symmetry (not icosahedral) / enveloped
- Hosts: various hosts; mammals (humans), insects, fish, and plants
- Replication:
 - occurs in cytoplasm (except some viruses infecting plants)
 - : especially, inside a specialized virus factory, called Negri body
 - viral mRNAs: having a 5’ cap and a 3’ polyadenylated tail (translated by cellular ribosomes)
 - encoding five proteins: L, P, G, N, and M proteins
 - L and P: regulation proteins
 - G and M: structural proteins
 - N: RNA-binding protein



Rhabdoviridae...

- Classification:
 - Genus Lyssavirus (Rabies virus, etc.)
 - etc.
- Rabies virus:
 - neurotropic virus: infection of the central nervous system
 - causing fatal disease in humans and animals (once symptoms appear)
 - transmitted through the saliva of infected animals
 - clinical signs:
 - furious form (광폭형): replication in the limbic system
 - dumb form (마비형): replication in the neocortex
 - flu-like symptoms, agitation, confusion, insomnia, paranoia, terror, hallucination, delirium, paralysis, etc.
 - diagnosis: RT-PCR, virus isolation, detection of Negri bodies, etc.
 - control
 - vaccine: prevention and therapy
 - no specific antiviral drugs, but immunoglobulin available



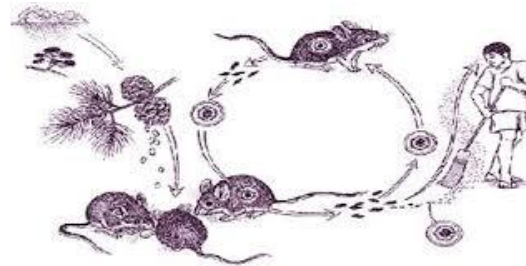
Bunyaviridae

Bunyaviridae... (currently, Bunyavirales)

- Etymology: first isolated in Bunyamwera, Uganda, Africa
- Enveloped; icosahedral
- Segmented genome: 3 segments (L, M, and S RNA)
 - L RNA: L protein (viral RDRP)
 - M RNA: viral glycoproteins (projecting from the viral surface / adding the virus in attaching to and entering the host cell
 - S RNA: nucleocapsid protein (N) and nonstructural protein (NS_s)
 - replication in cytoplasm
- Hosts (vectors): arthropods (mosquitos, tick, sandfly) or rodents → humans / can also infect plants
- Classification: 5 genera
 - Genus Hantavirus (currently, family *Hantaviridae*): *Orthohantavirus* (previously, *Hantaan virus*)
 - Genus Nairovirus (currently, family *Nairoviridae*): *Crimean–Congo hemorrhagic fever (orthonairo)virus*, etc.
 - etc.

Bunyaviridae...

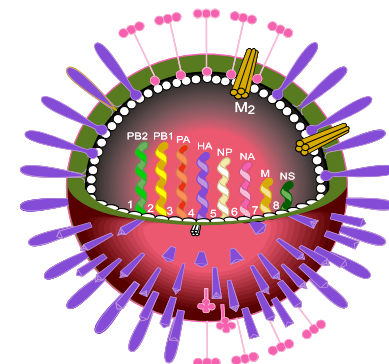
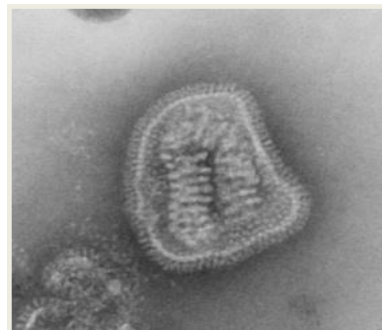
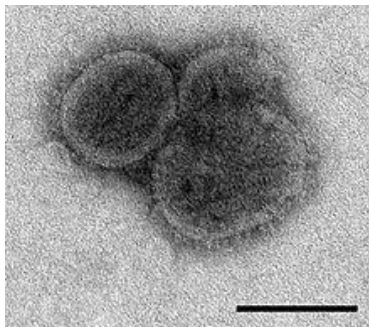
- (Ortho)hantavirus (or Hantan virus)
 - causing:
 - hemorrhagic fever with renal syndrome (in Asia and Europe)
(신증후출혈열; previously, Korean hemorrhagic fever)
: fever, lethargy, headache, rash, hemorrhage, anemia, thrombopenia, albuminuria, etc.
 - hantavirus (cardio)pulmonary syndrome (한타바이러스 폐증후군) (in Americas)
: initial symptoms similar as HFRS → respiratory symptoms (coughing, shortness of breath, elevated heart rate, impairment of lymphoid organs, cardiovascular shock, etc.
 - mortality rate of 7%
 - first isolated at Hantan River by HW Lee in 1976
 - transmission: bodily fluids (particularly, saliva from bites) of rodents and inhalation of virus particles from urine and feces (*Apodemus agrarius* and others)
 - diagnosis: RT-PCR, ELISA, etc.
 - control: vaccine (Hantavax in Korea and China, not in the US) / ribavirin



Orthomyxoviridae

Class 5: (-) ssRNA viruses (Orthomyxoviridae)

- Etymology: G. *orthos* = “straight”; G. *myxa* = “mucus”
- Enveloped / (–) ss RNA; segmented
- Six genera: A, B, C, D (Thogotovirus), Isavirus, and Quaranjavirus
 - A, B and C: viruses causing **influenza** in vertebrates
 - Thogotovirus: arboviruses (transmitted by arthropods, usually ticks) / infecting mammals or birds
 - Isavirus: infecting salmon
 - Quaranjavirus: arboviruses or infecting birds
- Influenza: Latin *influentia* = “epidemic”



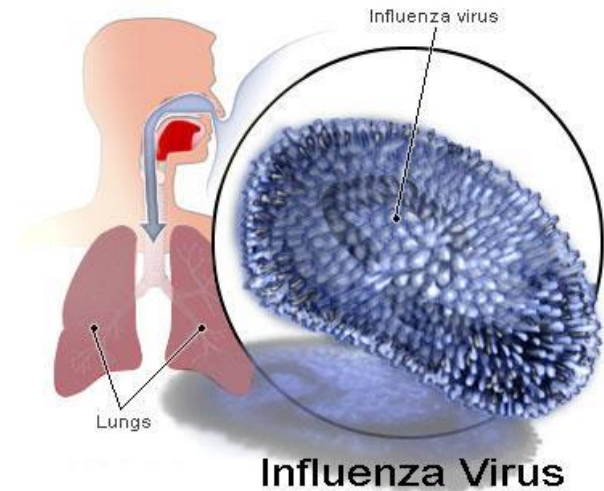
Common cold vs. Flu

▪ Common cold

- Rhinovirus
- Coronavirus
- Influenza virus
- Respiratory syncytial virus
- Adenovirus
- Bacteria: *Chlamydia pneumoniae*, *Haemophilus influenzae*, *Streptococcus pneumoniae*, *Mycoplasma pneumoniae*, etc.

▪ Influenza (Flu)

- Influenza virus
- *Influenza is an infectious disease caused by influenza virus.*

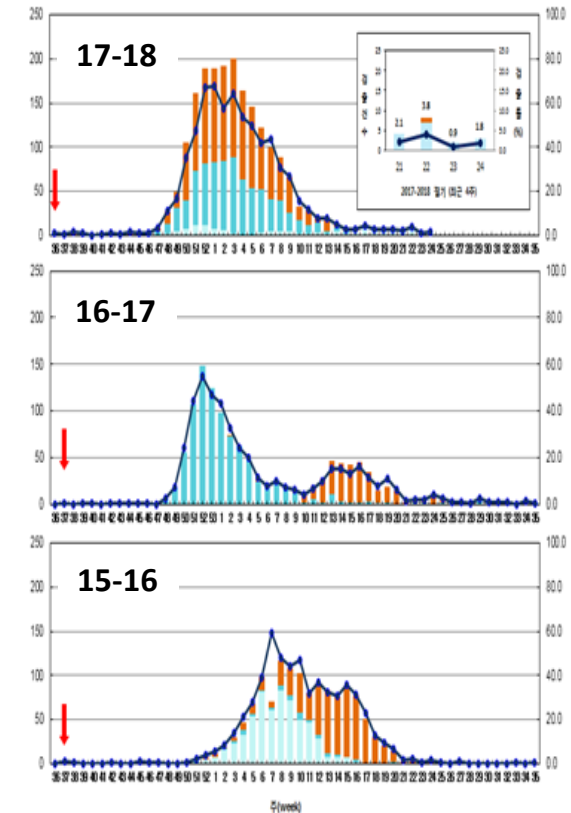


A, B and C influenza viruses

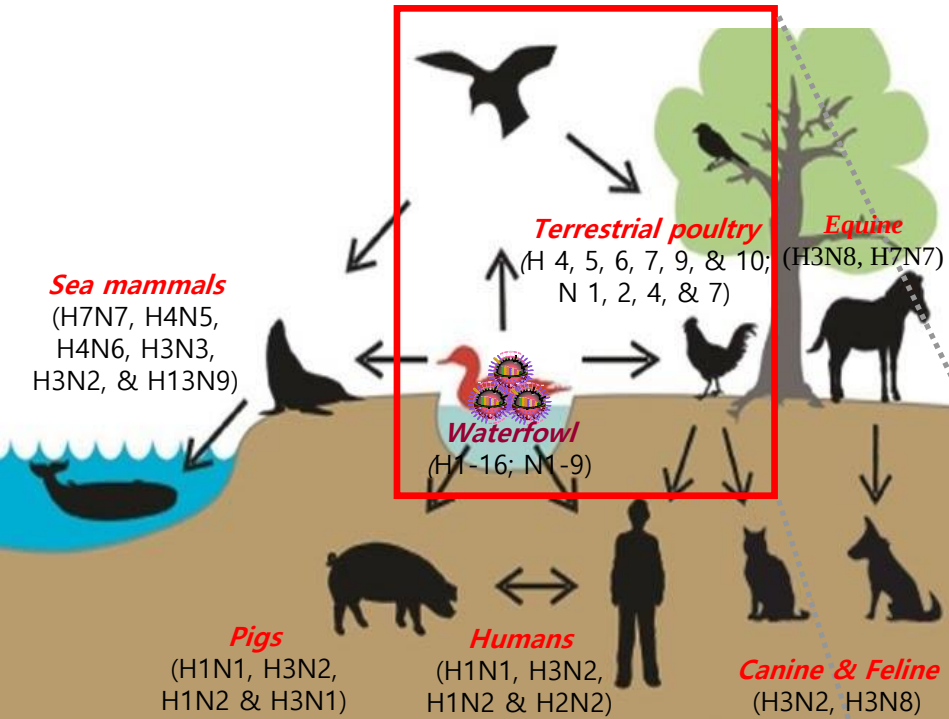
- ❖ The family ***Orthomyxoviridae*** : genera (types) A, B and C / eight or seven -segmented genome (ss (-) RNA) / enveloped
- ❖ **Influenza A**: further classified into various subtypes based on two proteins on the virus surface hemagglutinin (HA or H) and neuraminidase (NA or N) (**18 HA** and **11 NA subtypes** – e.g., H1N1 and H3N2) / infecting **various hosts** (including birds and mammals)
- ❖ **Influenza B**: classified into two antigenically distinct lineages, **Yamagata** and **Victoria** (-like viruses) / infecting **humans, seals and ferrets**
- ❖ **Influenza C**: infecting **humans, pigs and dogs**
- ❖ Influenza A and B viruses cause **seasonal epidemics**, while type C viruses cause mild respiratory illness.

Human seasonal influenza

- ❖ **Influenza epidemic:** emerges during each winter's flu season
- ❖ **Symptoms:** sudden onset starting with a headache, chills and cough, followed by rapidly **fever**, loss of appetite, muscle aches and fatigue, runny nose, sneezing, watery eyes and throat irritation
- ❖ **Deaths: 250,000 to 500,000 people each year worldwide:** mostly, occur in adults aged <65
- ❖ **Causative agents:** A/H1N1, A/H3N2, B influenza virus



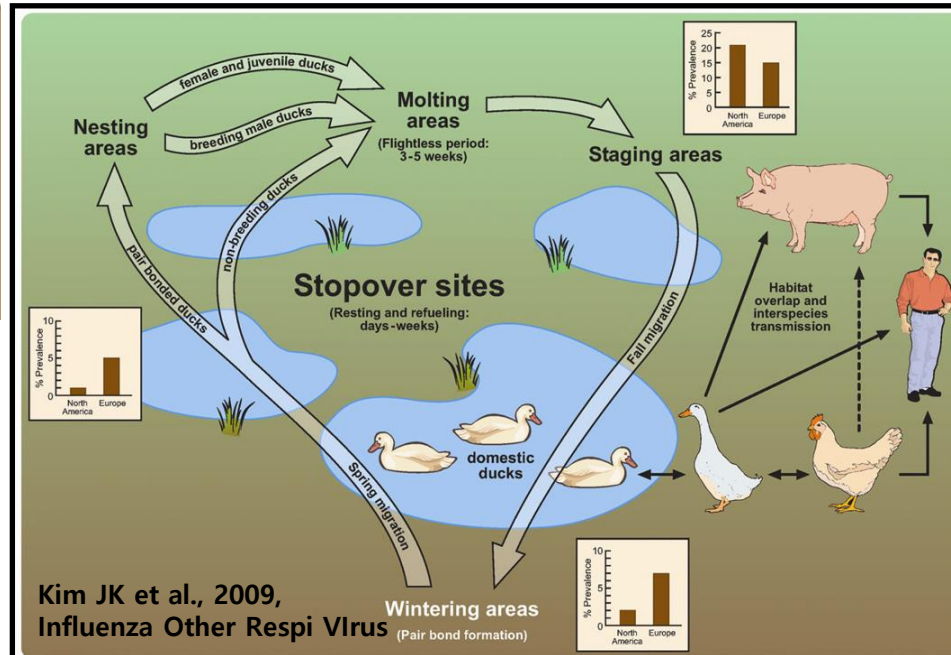
Class 5: (-) ssRNA viruses (Orthomyxoviridae)



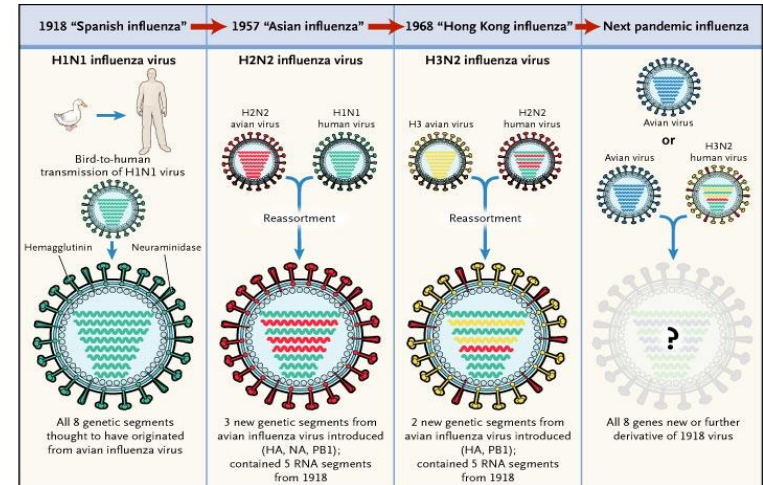
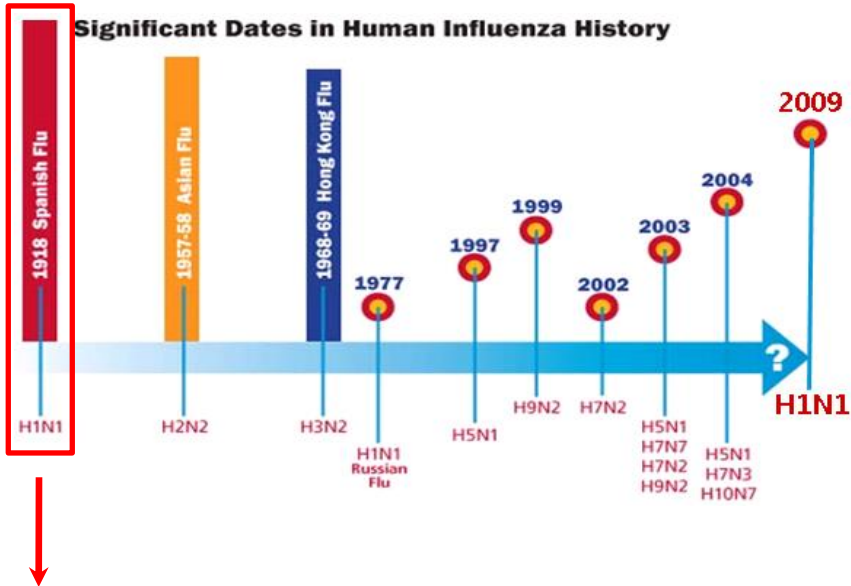
❖ INFLUENZA A VIRUS

- Enveloped, ss (-) RNA virus
- 8-segmented genome
: 2 surface + 6 internal genes
- 18 HA & 11 NA subtypes
- Host: various animals
- Natural host: no or Low pathogenic

자료원: **Fields Vriology (2007) 5th edition**, Knipe, DM & Howley, PM, eds, Wolters Kluwer/ Lippincott Williams & Wilkins, Philadelphia, Fig 48.1

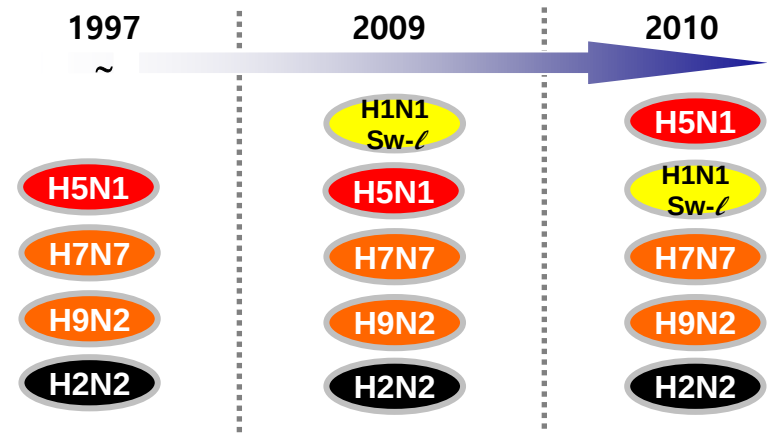


Type A / flu pandemics



- The most **notorious** pandemic influenza
- Mortality rate: unknown, but **10-20%**
- **~100 million people** were estimated to be killed.

The possible pandemic "Hit List"





Controlling for animals...

- ❖ **Traditional methods:** quarantine, restricted movements, culling of infected animals followed by eradication, etc.
- ❖ **Advanced methods:** vaccination
→ mainly, inactivated whole virus vaccines
- ❖ **Allopathy (symptomatic treatments):**
Companion or high-priced animals



Controlling for humans...

❖ Vaccination:

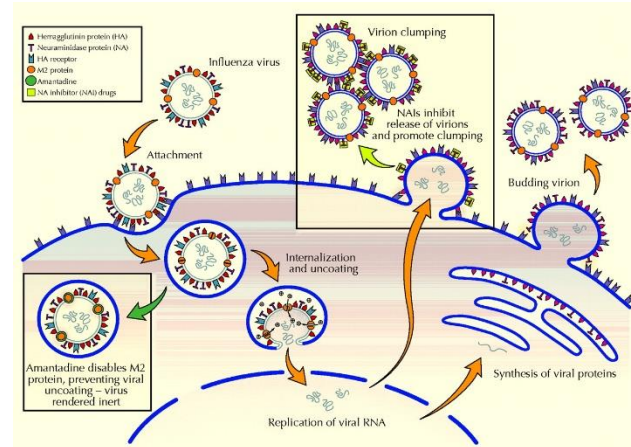
- mainly, split vaccines
- Live-attenuated vaccines
- tri- or quadrivalent

: A/California/7/2009 (H1N1) + A/Texas/50/ 2012 (H3N2) + B/Massachusetts/02/2012 (ex: 2014-2015 season)

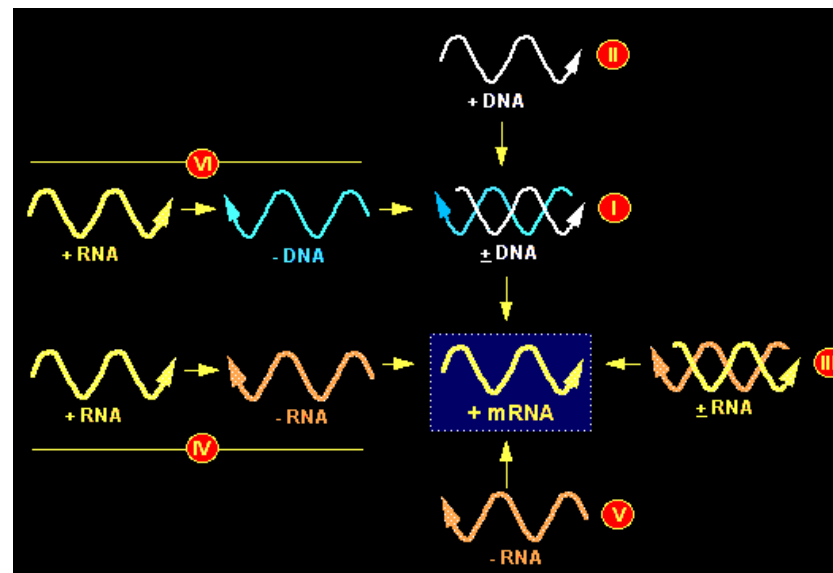
❖ Antiviral drugs:

- NA inhibitors: Oseltamivir, Zanamivir
- M2 ion channel inhibitors: Amantadine, Rimantadine
- Baloxavir (PA), Favipiravir (RDRP), Ingavirin (NP), Umifenovir (Arbidol; memb. fusion), etc.

❖ Allopathy (symptomatic treatments)



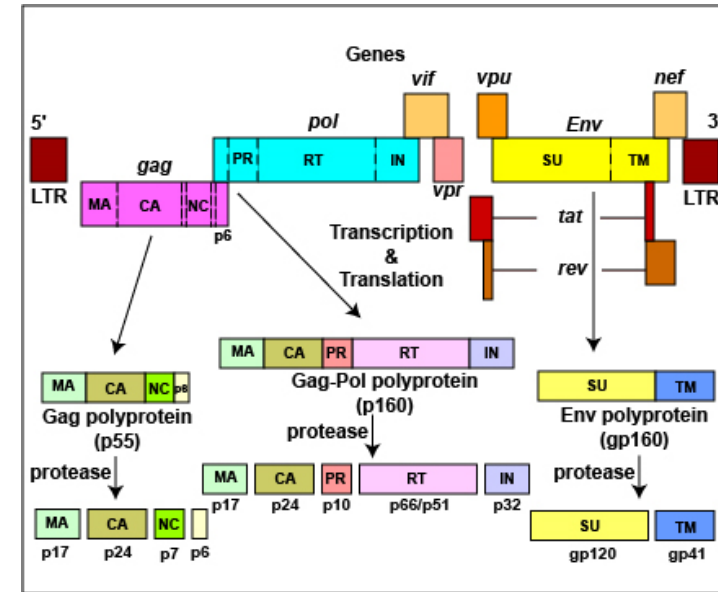
Group 6: +ssRNA-RT virus



- Family *Metaviridae*
- Family *Pseudoviridae*
- Family *Retroviridae* (Retroviruses)
 - Subfamily *Orthoretrovirinae*: Alpha-, Beta-, Delta-, Epsilon-, Gammaretrovirus, Lentivirus
 - Subfamily *Spumaretrovirinae*: Spumavirus

❖ General characteristics

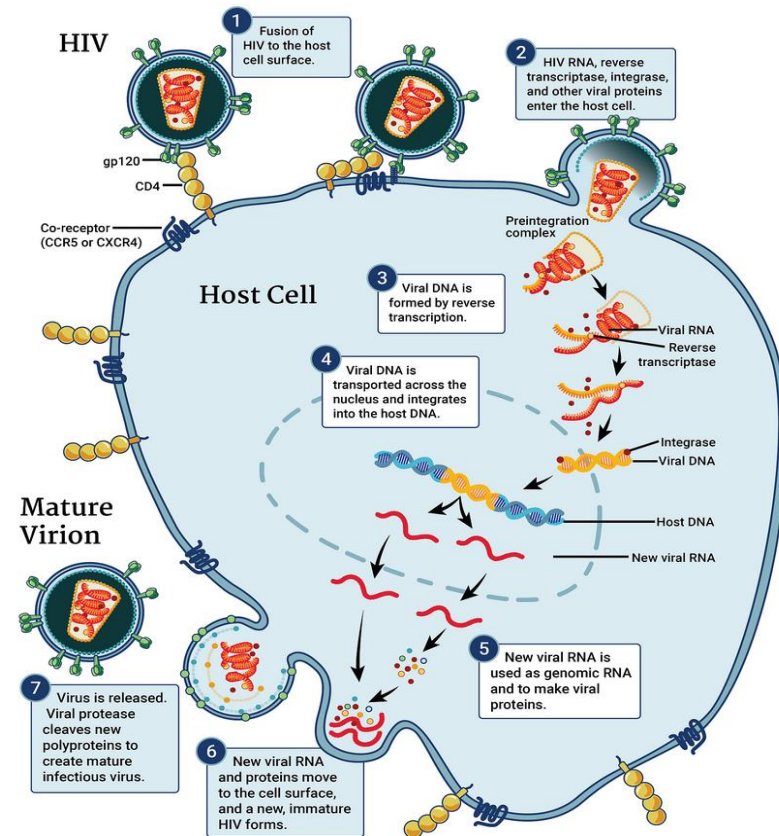
- Etymology: “retro-” = backwards
- Structure
 - Size: about 100 nm in diameter
 - Envelope:
 - lipid: obtained from the host plasma memb.
 - glycoprotein: encoded by the env gene
 - Genome:
 - dimer of (+) single-stranded RNA
 - : (+) ss RNA → (–) DNA → (±) DNA → (+) mRNA → Proteins
 - Genes and proteins:
 - gag: coding core and structural proteins (e.g. viral capsid)
 - pol: coding for reverse transcriptase, integrase, and protease
 - protease: proteolytic cleavage to make mature gag and pol proteins
 - env: coding for the viral coat (envelope) proteins
 - enabling the retrovirus to enter/exit host cells through endosomal membrane trafficking
 - protection from the extracellular environment via the lipid bilayer
 - the ability to enter cells: receptor-binding proteins



❖ General characteristics

• Life cycle

- **Attachment:** binding of the env protein to the receptor protein
- **Penetration and uncoating:** entrance of the RNA strands and enzymes (reverse transcriptase, integrase and protease) into the cell
- **Reverse transcription:**
 - (+) ss RNA $\rightarrow$ (-) DNA $\rightarrow$ ($\pm$) DNA $\rightarrow$ (+) mRNA $\rightarrow$ Proteins
 - : generation of cDNA from an RNA template
 - viral reverse transcriptase: RNA-dependent DNA polymerase
- **Generation of dsDNA:**
 - (+) ss RNA $\rightarrow$ (-) DNA $\rightarrow$ ($\pm$) DNA $\rightarrow$ (+) mRNA $\rightarrow$ Proteins
- **Integration:**
 - incorporation of dsDNA into the host chromosomal DNA $\rightarrow$ “provirus”
 - viral integrase
- Genome replication/protein synthesis
- Assembly and Release



❖ Classification (family *Retroviridae*)

- Genus **Alpharetrovirus**
 - causing sarcomas, other tumors, and anaemia of wild and domestic birds and rats
 - Rous sarcoma virus (RSV), etc.
 - the first oncogenic retrovirus / causing sarcoma in chickens
 - discovered in 1911 by Peyton Rous
- Genus **Betaretrovirus**
 - causing tumors in mice, monkeys, and sheep
- Genus **Gammaretrovirus**
 - causing leukemia or sarcoma in mice, cats, and monkeys
- Genus **Deltaretrovirus**
 - causing leukemia in cows, monkeys, and humans
 - Human T-lymphotropic virus (HTLV; 사람 T 세포 백혈병 바이러스)
- Genus **Epsilonretrovirus**
 - causing dermal sarcoma in fish
- Genus **Lentivirus**
 - Human immunodeficiency virus (HIV), SIV, FIV, etc.
- Genus **Spumavirus**
 - infecting cats, cows, horses, monkeys, and humans
 - virus' etiological role is unclear.

❖ Classification (family *Retroviridae*)

- Genus **Deltaretrovirus**

- : **Human T-lymphotropic virus** (HTLV; 사람 T 세포 백혈병 바이러스)

- Four types: HTLA-I, -II, -III, and -IV

- Causing cancer (adult T-cell leukemia and lymphoma, ATLL; 성체 T 세포 백혈병) and demyelinating disease (HTLA-1 associated myelopathy/Tropical spastic paraparesis; HAM/TSP)

- Transmission: sexually transmitted / breast feeding by the carrier

- Clinical signs:

- mainly asymptomatic infection

- ~5% of infection: causing ATLL (after 30 to 50 yrs) → death (in 1 yr)

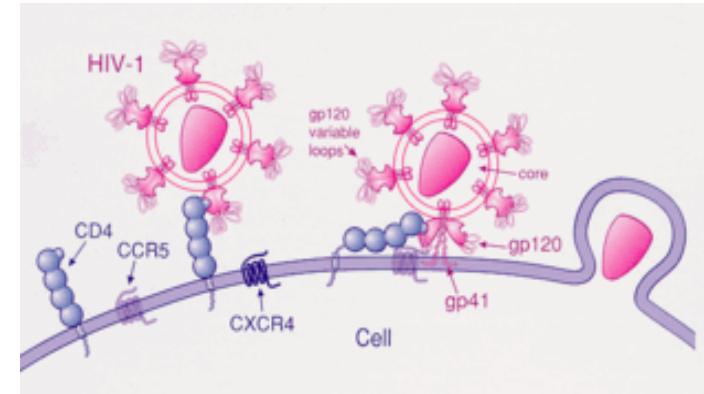
- Distribution: southern Japan, Caribbean, Central Africa, southeast USA, etc.

- Diagnosis: Ag or Ab detection (e.g. ELISA), viral RNA detection (e.g. PCR), etc.

- Therapy: no specific treatment, but Azidothymidine (AZT; Zidovudine/Retrovir)+INF- α is partially effective

❖ Classification (family *Retroviridae*)

- Genus **Lentivirus** (Lenti=slowly replicating)
 - : **Human immunodeficiency virus (HIV)**
 - Acquired immunodeficiency syndrome (AIDS)
 - : progressive failure of the immune system → life-threatening opportunistic infections and cancers
 - Two major types: HIV-1 (high infectivity and virulence; global prevalence) / HIV-2 (low infectivity and virulence; west Africa)
 - Groups/Subtypes:
 - HIV-1: groups M (“major”; subtypes A to L), N, O and possibly P
 - HIV-2: groups A to H
 - Host cells: CD4+ immune cells (helper T cells, macrophages, etc.)
 - Tropism: macrophage (M)-tropic and T cell (T)-tropic
 - Receptor: CD4 vs. env protein (gp160 → gp120 + gp41)
 - Co-receptor: CCR5 (M-tropic) or CXCR4 (T-tropic)



❖ Classification (family *Retroviridae*)

- Genus **Lentivirus** (Lenti=slowly replicating)
 - : **Human immunodeficiency virus (HIV)**
 - Transmission: sexual contact (semen, vaginal fluid), transfusion (blood), or breast milk
 - Clinical signs:
 - early stage (HIV infection): fever, malaise, headache, skin rash, weight loss, adenitis, oral candidosis, etc. → producing HIV Ab
 - late stage (AIDS; 8–10 yrs after HIV infection)
 - opportunistic infections: candidosis, Herpesvirus (HCMV, HSV) infection, Pneumonia, etc.
→ see p. 588 (표 10-21 AIDS와 연관된 질병), 약품미생물학 분과 교재
 - central nervous system diseases: ataxia, dementia, autoimmune neuropathy, cerebrovascular disease, brain tumor, etc.
 - cancers: Kaposi's sarcoma, oral cancer, rectal cancer, B-cell lymphoma, etc.

Class 6: (+) ssRNA-RT viruses (HIV)

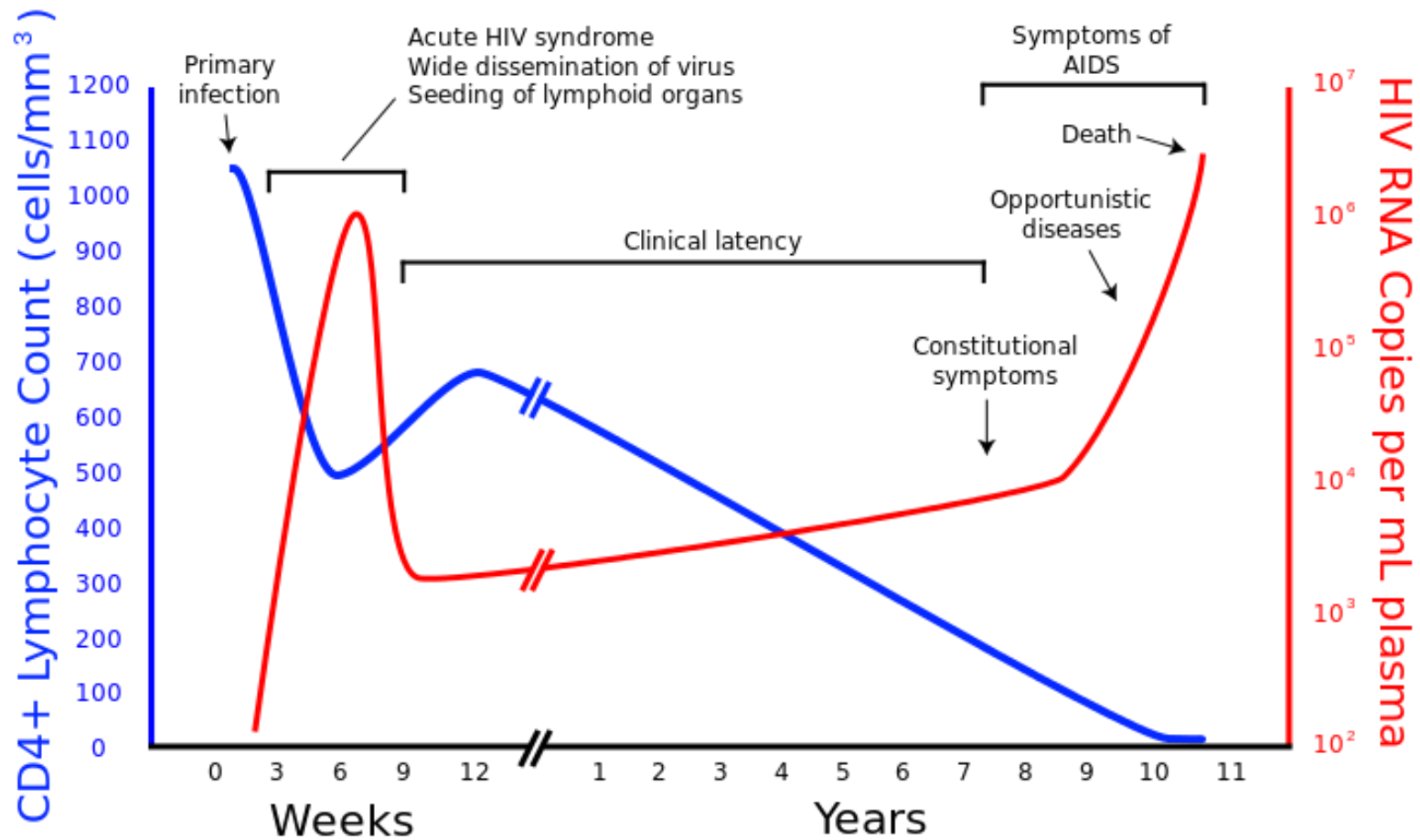


표 10-21 AIDS와 연관된 질병

기관지, 기관, 폐와 칸디다증
식도, 칸디다증
침윤성 자궁경부암
전신성 또는 폐 외의 콕시디오이레스 진균증
만성 장 크립토스포리디아증(cryptosporidiosis, 1달 이상 간)
Cyclospora, 설사병
거대세포 바이러스 질병(간, 비장이나 림프절 이외의)
거대세포 바이러스 망막염(시력상실과 함께)
HIV 관련 뇌병증
단순포진: 만성궤양(1달 이상 간);기관지염, 폐렴, 식도염
전신성 또는 폐외 히스토플라스마증
만성 장 이소스포라증(Isosporiasis, 1달 이상 간)
카포시육종
버킷림프종(또는 equivalent term)
면역 모세포의 림프종(또는 equivalent term)
뇌의 1차 림프종
<i>Mycobacterium avium complex</i> 또는 <i>M. kansasii</i>
결핵, 어느 부위든지 가능
다른 종이나 동정되지 않은 <i>Mycobacterium</i>
<i>Pneumocystis carinii</i> 폐렴
재발성 폐렴
진행성 다초점성 백질의 병증
재발성 살모넬라 패혈증
뇌의 톡소플라스마증
AIDS로 인한 소모성 증후군

❖ Classification (family *Retroviridae*)

- Genus *Lentivirus* (Lenti=slowly replicating)
 - : Human immunodeficiency virus (HIV)
 - **Diagnosis:** HIV-specific Ab detection (ELISA), indirect immunofluorescence staining, PCR / counting virus particles and CD4+ T cells in blood
 - **Therapy:**
 - Nucleoside Analogue Reverse Transcriptase Inhibitors (NRTI)
 - Non-nucleoside Reverse Transcriptase Inhibitors (NNRTI)
 - Protease Inhibitor (PI)
 - Attachment/Fusion Inhibitors
 - Integrase inhibitors
 - Cocktail (HAART):
 - see p. 589 (표 10-22 항 HIV 제제), 약품미생물학 분과 교재

표 10-22 항 HIV 제제

뉴클레오시드 유사 역전사효소 억제제 (Nucleoside Analogue Reverse Transcriptase Inhibitors, NRTI)

Azidothymidine(AZT) (Zidovudine/Retrovir)
Dideoxycytidine(ddC) (Zalcitabine)
Dideoxyinosine(ddI) (Didanosine)
d4T (Stavudine)
3TC (Lamivudine)
Tenofovir disoproxil fumarate (adenosine class) (Viread)
ABC (Abacavir)
FTC (Emtricitabine (Emtriva))

비뉴클레오시드 역전사효소 억제제 (Non-nucleoside Reverse Transcriptase Inhibitors, NNRTI)

Nevirapine (Viramune)
Delavirdine (Rescriptor)
Efavirenz (Sustiva)

단백질분해효소 억제제 (Protease Inhibitors, PI)

Saquinavir
Tipranavir
Darunavir
Ritonavir (Norvir)
Indinavir (Crixivan)
Lopinavir (Kaletra)
Nelfinavir (Viracept)
Amprenavir (Agenerase)
Fosamprenavir (Lexiva)
Atazanavir (Reyataz)
Tipranavir (Aptivus)

결합 및 세포막융합 억제제

CCR5 Inhibitors (maraviroc)
T-20 (enfuvirtide/Fuzeon)

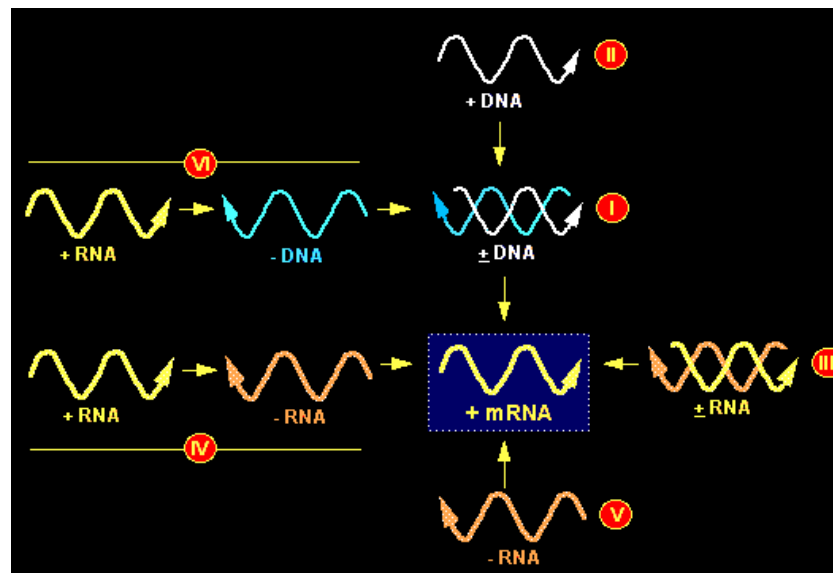
삽입효소 억제제 (integrase inhibitors)

Raltegravir (isentress)

각테일요법 (HAART)의 예

Abacavir/zidovudine/lamivudine (Trizivir)
NNRTI + 2 NRTI
Efavirenz + tenofovir + emtricitabine
Nevirapine + abacavir + lamivudine
2 PI + 2 NRTI
Atazanavir + ritonavir + zidovudine + lamivudine
Efavirenz + emtricitabine + tenofovir disoproxil fumarate (Atripla)

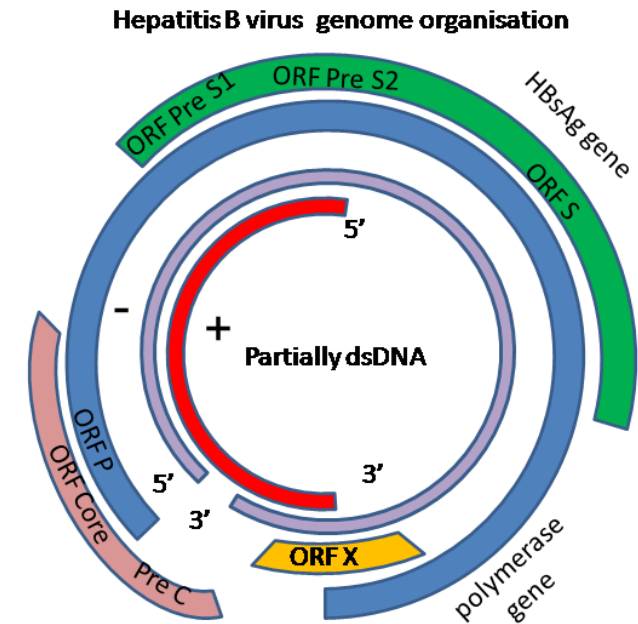
Group 7: dsDNA-RT virus



- Family *Hepadnaviridae*
 - Genus *Orthohepadnavirus*: infecting some mammals including humans (hepatitis B virus)
 - Genus *Avihepadnavirus*: infecting birds
- Family *Caulimoviridae*: plant viruses

❖ Hepatitis B virus (HBV)

- One of the **smallest** animal virus
 - 42–47 nm in diameter (complete virion; Dane particle – infectious)
 - **HBsAg particle**:
 - virion particle without DNA
 - : ~22 nm in diameter
 - not infectious
 - **Envelope**: lipid + embedded proteins
 - **Icosahedral nucleocapsid** (protein):
 - encloses the viral DNA and DNA polymerase
 - DNA polymerase: reverse transcriptase (RT) activity
 - **Genome structure**
 - Partially double-stranded, circular DNA
 - ds DNA → pregenomic RNA intermediate: (+) ssRNA
 - pregenomic RNA:
 - acts as the mRNA for the translation
 - acts as the template for genome replication
 - : (+) ssRNA → (–) ssDNA → dsDNA
-



❖ Hepatitis B virus (HBV)

- **Clinical signs**

- early stage: fever, anorexia, fatigue, diarrhea, nausea, vomiting, etc.
→ most people: getting better within a couple of months
- some people:
 - 5~10% of adults; 30~50% of children; 80~90% of infants
→ developing chronic hepatitis B
 - 15~20% of chronic hepatitis B patients → death by liver cirrhosis or cancer

- **Diagnosis:** Ag or Ab–detecting ELISA

- **Prevention and therapy:**

- prevention: vaccine (recombinant HBsAg)
- therapy:
 - acute infection:
 - no specific treatment, but...
 - hepatitis B immunoglobulin: <1 week post-infection or newborn baby
 - chronic infection: lamivudine (targeting polymerase), nucleoside analogues (famciclovir, adefovir dipivoxil), INF- α (+lamivudine)

